



Motor imagery EEG signal classification using minimally random convolutional kernel transform and hybrid deep learning

Jamal Hwaidi^a, Mohamed Chahine Ghanem^{b,c},*

^a Department of Engineering, City St George, University of London, EC1V 0HB, London, UK

^b School of Computer Science and Informatics, University of Liverpool, L69 3BX, Liverpool, UK

^c School of Computer Science and Mathematics, Keele University, ST5 5BG, Keele, UK

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ABSTRACT

The brain-computer interface (BCI) establishes a non-muscle channel that enables direct communication between the human body and an external device. Electroencephalography (EEG) is a popular non-invasive technique for recording brain signals. It is critical to process and comprehend the hidden patterns linked to a specific cognitive or motor task, for instance, measured through the motor imagery brain-computer interface (MI-BCI). A significant challenge is presented by classifying motor imagery-based electroencephalogram (MI-EEG) tasks, given that EEG signals exhibit nonstationarity, time-variance, and individual diversity. Achieving good classification accuracy is also challenging due to the increasing number of classes and the inherent variability among individuals. To overcome these issues, this paper proposes a novel method for classifying EEG motor imagery signals that efficiently extracts features using the Minimally Random Convolutional Kernel Transform (MiniRocket). A linear classifier then utilises the extracted features for activity recognition. Furthermore, a novel deep learning model based on Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) architecture was proposed and demonstrated to serve as a baseline. The classification via MiniRocket's features achieved higher performance than the best deep learning models at a lower computational cost. PhysioNet and BCI Comp IV 2a datasets were used to evaluate the performance of the proposed approaches. Using PhysioNet, the proposed models achieved mean accuracy values of 98.63% and 98.06%, respectively, for the MiniRocket and CNN-LSTM. With the BCI-CompIV-2a dataset, proposed models achieved mean accuracy values of 92.57% and 92.32%, respectively. The findings demonstrate that the proposed approach can significantly enhance motor imagery EEG accuracy and provide new insights into the feature extraction and classification of MI-EEG. An additional future direction is non-additive electrode-source fusion (Choquet-integral/coalition formulations) to improve robustness under low-SNR EEG and inter-subject variability.

1. Introduction

A human-computer interaction technique based on brain signals is known as brain-computer interface (BCI) technology (Greenfield et al., 2012; Wolpaw et al., 2000; Moore, 2003). It offers a communication channel for non-neuromuscular control and communication between the human brain and the outside world using a brain-computer interface without the use of muscles or the peripheral nervous system (Zander and Kothe, 2011; Ghanem et al., 2015).

Electroencephalography (EEG) signals represent electrical signals from the brain nerves in the BCI system. It serves as the system's foundation for signal processing as well (Mathiyazhagan and Devasena, 2025; Wang et al., 2020). The electrical signals produced by the brain's neurons during EEG brain rhythms are microvolts (He et al., 2018). EEG

uses affordable equipment and permits patient movement while recording. These are advantages over other non-invasive recording methods such as magnetoencephalography (MEG) and functional magnetic resonance imaging (fMRI), which require patients to remain stationary while using expensive, large-scale equipment (Millán et al., 2004).

Various EEG signal types have been employed as BCI control signals. The most common signals are P300 evoked potentials, steady-state visual evoked potentials (SSVEP), and motor imagery (MI). The power spectrum of various frequency bands can change for various movement tasks, reflecting neuronal firing pattern changes. Event-related synchronisation (ERS) and event-related desynchronisation (ERD) are two names for this phenomenon. The primary spectrums of ERS and

* Corresponding author at: School of Computer Science and Informatics, University of Liverpool, L69 3BX, Liverpool, UK.

E-mail addresses: jamal.hwaidi@city.ac.uk (J. Hwaidi), mohamed.chahine.ghanem@liverpool.ac.uk (M.C. Ghanem).

ERD in MI tasks are μ (8–14 Hz) and β (14–30 Hz) (Pfurtscheller and Da Silva, 1999).

Different kinds of motor or cognitive activities can be understood using EEG. The term “motor imagery” (MI) describes a subject’s ability to move their limbs mentally even though they are not being moved (Yuan and He, 2014).

An emerging area of biomedical applications is BCI based on EEG motor imagery (Pfurtscheller and Neuper, 2001). Clinically, MI-BCIs have progressed classification towards closed-loop therapeutic and assessment systems, particularly in post-stroke neurorehabilitation where decoded MI is coupled to contingent feedback (robotic orthoses or functional electrical stimulation) to drive Hebbian-like plasticity. In particular, sham-controlled clinical studies have reported that daily BMI/BCI training can improve motor outcomes compared to physiotherapy alone, and subsequent meta-analyses indicate an overall positive effect while highlighting heterogeneity and the need for standardised protocols, robust multi-session generalisation, and reduced calibration burden. These deployments impose constraints not captured by offline benchmarks: low-latency inference, stability across days/sessions, robustness to fatigue/medication-related variability, and safety monitoring in real-world environments. The emphasis on computationally efficient, reproducible feature transforms (as pursued here with MiniRocket) is therefore directly relevant to embedded or bedside clinical-grade MI-BCI implementations. The motor imagery classification problem can also be used to identify and separate various types of thinking or imagining (Tonin et al., 2011; Ang et al., 2012). Due to its advantages over other cerebral signals, researchers have given special consideration to the classification of EEG signals. Although a specialist could also perform this classification, doing so would not be as quick or accurate, so various machine learning and deep learning techniques have been used to automate the classification of MI (Yang et al., 2018).

Beyond purely algorithmic countermeasures (artefact rejection, domain adaptation, and robust temporal feature learning), robustness against EEG non-stationarity is increasingly pursued via hybrid (multi-modal) BCIs, most prominently EEG–fNIRS. EEG provides millisecond-scale electrophysiological dynamics but is sensitive to impedance drift, motion/EMG contamination, and environmental interference; fNIRS measures task-related haemodynamic responses (HbO/HbR) and is comparatively less affected by electromagnetic noise, offering complementary information at slower time scales. Cross-modal fusion can therefore act as an implicit regulariser under distribution shift by constraining the latent neural state with an additional physiological channel and by exposing confounding physiological-state factors (fatigue/hypoxia) that modulate EEG statistics. Recent open-access resources explicitly target such stress-condition shifts; for example, an EEG+fNIRS dataset collected over two weeks of high-altitude exposure during Stroop tasks was released to support algorithm development for oxygen-deficiency monitoring and robustness benchmarking under extreme environments. While the present work focuses on unimodal MI-EEG decoding, the proposed MiniRocket-based pipeline is directly extensible to multimodal settings by applying the transform per modality and fusing at feature- or decision-level (Ji et al., 2024).

Traditional MI-BCI classification approaches are generally categorised into two groups based on the features of EEG signals: (1) spatial feature classification (Hekmatmanesh et al., 2020) and (2) spatial-frequency feature classification (Thomas et al., 2009). According to Sakhavi et al. (2018), several prior MI-BCI algorithms neglected the temporal characteristics of EEG signals and neglected to consider the dynamic energy representation of EEG. Automatic function selectors are only partially effective at reversing this tendency.

Deep learning (DL) techniques have recently outperformed traditional handcrafted techniques in a number of fields, including image processing, speech processing, video processing, and text processing (de Haan et al., 2019; Fenil et al., 2019; Sahu and Anand, 2018). EEG signal characteristics like low signal-to-noise ratio (SNR), fewer data,

and multiple channels make it challenging to develop a general DL model for the identification of EEG signals.

It is important to have a meaningful design for the DL framework used to process EEG signals. Some successes have been seen in researchers’ efforts to apply DL techniques to the BCI field.

Deep neural networks (DNN) with more data, improved learning methods, and faster computation have gained in popularity (Bengio and Bengio, 1999). By processing the data from an EEG time series in the RNN networks’ internal memory, EEG signals can be decoded (Bengio et al., 1994). As opposed to a normal RNN, Long Short-Term Memory (LSTM) is an RNN with the capacity to maintain the sequence of data over an extended period and identify the desired pattern (Graves and Schmidhuber, 2005). Convolutional neural network (CNN) models can recognise strong spatial details in images (LeCun et al., 1998). Researchers have extensively applied to investigate the classification and spatial characteristics of EEG signals (Amin et al., 2019; Lawhern et al., 2018; Riyad et al., 2020). To improve DNNs’ capacity to simultaneously extract spatial and temporal characteristics, CNN and LSTM neural networks were combined to create a hybrid neural network that can learn both spatial and temporal features (Zhang et al., 2019).

Despite the impressive accomplishments of prior work in this field, the BCI system still lacks standards for practical application. Moreover, there is still considerable opportunity for improvement in both the EEG signals classification methodology and accuracy.

The main contributions of this paper are as follows:

- (1) The independent component analysis (ICA) technique is utilised to separate the μ and β frequencies as segmentation targets from other frequencies.
- (2) The MI EEG signals are subjected to a Minirocket feature extraction method. The MiniRocket can enhance the accuracy of EEG classification by retaining more crucial feature information while also processing less data.
- (3) A hybrid neural network, which consists of CNN and LSTM, is utilised to enhance motor imagery accuracy in EEG classification and compared to MiniRocket’s feature-based classifier.

Recent approaches to MI-BCI can be grouped into three broad categories: (i) deterministic time-series transforms such as ROCKET and MiniRocket that generate features for lightweight classifiers, (ii) compact CNN/LSTM hybrids that learn spatial and temporal features end-to-end, and (iii) light-weight classifiers like ridge regression or SVM. Our work focuses on a systematic comparison between a deterministic transform (MiniRocket + ridge) and a compact CNN–LSTM network, quantifying both accuracy and computational cost. The remainder of the paper is organised as follows: Section 2 reviews the literature on the classification of EEG MI utilising machine learning and deep learning techniques. Section 3 presents the data description and system method. The discussion of the methodology with existing work and conclusion are given in Sections 4 and 5, respectively. For clarity we summarise related MI-BCI literature next (Section 2) focusing on methods directly compared to our pipeline (MiniRocket + linear classifier and compact CNN–LSTM), rather than a broad survey.

2. Related work

We focus this short related-work summary on MI-EEG methods directly comparable to our contribution: (i) time-series transforms (ROCKET/MiniRocket family), (ii) compact CNN/LSTM hybrids, and (iii) light-weight classifiers (ridge/SVM/LR). This concentrated review emphasises the central argument: MiniRocket offers near-SOTA accuracy with far fewer trainable parameters and lower CPU cost, while hybrid CNN–LSTM models provide an end-to-end baseline for spatio-temporal feature learning. Substantial efforts have been made in the past to improve the accuracy of the MI classification through feature extraction and classification algorithms (Lotte et al., 2018; Zhang et al., 2018).

Hybrid deep-learning models combining convolutional and recurrent layers have recently shown strong performance in MI-EEG classification. For example, [Bian et al. \(2024\)](#) reports a CNN–LSTM model augmented with generative adversarial networks to achieve 96.06. [H. Li et al. \(2022\)](#), [Khademi et al. \(2022\)](#) propose transfer-learning based CNN–LSTM and CNN–GRU hybrids, respectively, to leverage pretrained spatial filters. These works typically employ deeper architectures and data augmentation. In contrast, our CNN–LSTM baseline is deliberately compact and uses no generative augmentation, enabling a transparent comparison with a deterministic MiniRocket pipeline. To classify MI-EEG data, conventional machine learning techniques have been extensively used. The MI-EEG signal is typically processed using traditional methods in three steps: preprocessing, feature extraction, and classification.

Preprocessing involves a number of operations, such as channel selection, signal filtering, signal normalisation, and removal of artefacts (removing the noise from MI-EEG signals). Independent component analysis (ICA) is the technique that is most frequently used to remove artefacts ([Jafarifarmand and Badamchizadeh, 2019](#)).

Existing research indicates that a variety of feature extraction algorithms have been proposed for extracting task-related MI features from EEG signals with high dimensions. Depending on the processing domain for the data, the MI features can be categorised into three types: temporal features, spectral features, and spatial features. Temporal features like mean, variance, Hjorth parameters, and skewness are derived from the time domain at various time points or throughout various time segments ([Pawar and Dhage, 2020](#)). Spectral features can be either time-frequency features like short-time Fourier transform (STFT) ([Hyvärinen et al., 2010](#)) and wavelet transform (WT) ([Zhou et al., 2016](#); [Malan and Sharma, 2022](#)) or frequency-domain features like power spectral density (PSD) and fast Fourier transform (FFT) ([Djamal et al., 2017](#)). Spatial features, such as common spatial patterns (CSP), are intended to identify characteristics from specific scalp electrode locations ([Ramoser et al., 2000](#); [Wang et al., 2006](#)). The regularisation feature in sparse CSP ([Fu et al., 2020](#)) adds sparsity to CSP values. Other methods that have been proposed to improve CSP functionality include stationary CSP ([Samek et al., 2012](#)), divergence CSP ([Samek et al., 2013](#)), sub-band CSP (SBCSP) ([Novi et al., 2007](#)), probabilistic CSP ([Wu et al., 2014](#)), and frequency-domain CSP (FDCSP) ([Wang et al., 2017](#)). Filter bank CSP (FBCSP) ([Ang et al., 2008](#)) is an enhanced variation of the CSP approach that leverages frequency information in MI-EEG signals and spatial information across EEG channels. Furthermore, discriminative filter bank CSP (DFBCSP) ([Higashi and Tanaka, 2012](#)) is an extension of CSP that uses optimisation to create spatial weights for finite impulse response filters.

During the classification phase, various classifiers were used, for instance, the naive Bayesian classifier ([Ang et al., 2012](#)), linear discriminant analysis (LDA) ([Chen et al., 2014](#)), support vector machine (SVM) ([Chatterjee and Bandyopadhyay, 2016](#)), K- nearest-neighbour (KNN) ([Hu et al., 2016](#)), and extreme learning machine (ELM) ([Wang et al., 2019](#); [Zhang et al., 2020](#)), to categorise the derived MI features into different MI tasks.

[Malan and Sharma \(2022\)](#) used a neighbourhood component analysis (NCA)-based supervised learning framework to enhance the spatial features that CSP had obtained for each sub-band. SVM was ultimately trained to handle the classification. SVM is inefficient when there are a large number of observation samples and is sensitive to missing data, despite the fact that it can perform inference in a reasonably quick amount of time. KNN's use in classifying MI EEG signals is constrained by its high time and space complexity, which is not sensitive to missing values. Generally, even though the traditional machine learning algorithms have produced impressive results, their accuracy has not been satisfactory.

On the other hand, being able to automatically extract the discriminative features, deep learning methods with deep neural networks (DNN) have recently performed remarkably well in several areas,

such as speech recognition, computer vision, and medical diagnosis ([Qamhan et al., 2021](#); [Altaheri et al., 2019](#); [Zhao et al., 2019](#); [Han et al., 2022](#); [Muhammad et al., 2020, 2017](#)). The growing popularity of deep learning has significantly reduced the need for feature extraction. Deep learning algorithms can produce and predict new features without specifying which ones to use or how to extract them ([Sun et al., 2017](#); [Mahajan et al., 2022](#)).

To classify MI EEG, various researchers have used deep learning methods. CNN can learn reliable spatial features ([Schirrmester et al., 2017](#)). RNN can successfully learn the EEG signal's sequential relationship ([Luo et al., 2018](#)). For unsupervised feature learning, autoencoder (AE) models are appropriate ([Bozhkov and Georgieva, 2020](#)). Shallow CNN (sCNN) and deep CNN (dCNN) architectures were developed by [Schirrmester et al. \(2017\)](#) to extract a wide range of features and to comprehend the temporal structure of band power variations. However, the dCNN underperformed the sCNN because there was less training data available. Studies have indicated that the CNN's various layers can extract various abstractions from the EEG signal. As a result, [Amin et al. \(2019\)](#) layered and filtered extracted features from CNN models to discover how time-series signals are structured both locally and overall. The classification accuracy was greatly improved by combining features. Moreover, the pragmatic CNN (pCNN) network was developed by [Tayeb et al. \(2019\)](#), which is less complicated and more accurate than dCNN. Tabar and Halici ([Tabar and Halici, 2016](#)) developed a network to classify two classes of MI tasks using CNN and stacked autoencoders (SAE). They first used Short-Time Fourier Transform (STFT) to extract information from the EEG signal's time and frequency domains and convert it into two-dimensional (2D) images, forwarding these images to the CNN-SAE network for classification. To improve differential phase-synchronised representations for examining the EEG-synchronised transitions from being alert to being drowsy, [Reddy et al. \(2019\)](#) used fuzzy common-space patterns. The findings of the data analysis indicate that traditional regression methods like support vector regression (SVR) and ridge regression are outperformed by the suggested multi-task deep network. Four expert CNNs were combined into a modular network by [Olivas-Padilla and Chacon-Murguia \(2019\)](#). Binary classification is carried out by each expert CNN, and the results are then fully connectedly analysed.

Recent MI studies have focused on multiscale spatial–temporal fusion and extremely efficient time-series transforms. Notably, Jin et al. in [Jin et al. \(2025\)](#) present a multiscale spatial–temporal fusion network for MI, and the MiniRocket transform ([Dempster et al., 2020](#)) by Dempster et al. that has been demonstrated as a very fast, high-accuracy transform for general time-series classification; these works motivated our combined MiniRocket vs CNN–LSTM comparison and the computational benchmarking we now report

EEG signals are time-sequenced nonlinear signals. Time series with nonlinear characteristics, like EEG, are frequently handled by long-short-term memory (LSTM), a RNN variant. The LSTM network was employed for the first time by [Shen et al. \(2017\)](#) to deal with issues with EEG signal classification, and they achieved notable improvements in both the MI-EEG dataset used in BCI competitions with individuals who are healthy, as well as the dataset compiled from stroke patients. Speech recognition and natural language processing were the inspirations for this work. The LSTM deep neural network was employed by [Tortora et al. \(2020\)](#) in order to deal with time-dependent information contained in brain signals during locomotion. [Lu et al. \(2019\)](#) classified sequential-spatial-frequency features using a deep RNN architecture. Additionally, by viewing the signal as having various non-stationary and nonlinear characteristics, this method also reduces variations in classification between individuals. [Ma et al. \(2018\)](#) extracted both spatial and temporal properties from the *eeegnmdb* dataset by combining the LSTM and bidirectional LSTM (BiLSTM), obtaining a result that was 8.25% more accurate than conventional methods.

Recent advances have explored multiscale spatial–temporal fusion and efficient time-series transforms for MI EEG decoding. For example,

a multiscale spatial–temporal feature fusion neural network (MSTFNet) was recently proposed for MI-BCI and shows state-of-the-art performance on several benchmarks (Jin et al., 2021) and Jin et al. (2025). In parallel, the ROCKET family (ROCKET → MINIROCKET) has been developed specifically for fast, highly accurate time-series transforms and has been applied across many domains; MINIROCKET offers near-ROCKET accuracy with substantial speedups. Several recent papers apply MiniRocket/ROCKET variants to physiological signals or propose Rocket ensembles for multivariate EEG/MEG. These developments motivate our focused comparison between a MiniRocket+linear pipeline and a compact CNN-LSTM baseline for MI EEG classification (Jin et al., 2025).

Extracting both spatial and temporal information simultaneously, a variety of neural networks have been combined (hybrid DNN). Zhang et al. (2021) employed CNN and LSTM to extract the signal's sequence relationship and frequency-spatial information. Due to their ability to simultaneously learn multiple features, hybrid neural networks have been shown to outperform other neural network architectures. H. Li et al. (2022) proposed a CNN- and LSTM-based multilevel spatial–temporal feature fusion EEG classification algorithm. Integral extraction and fusion of the middle layer feature, the spatial feature, and the temporal feature. It overcomes the traditional machine learning algorithms' tendency to easily lose manually selected features. Furthermore, Khademi et al. (2022), Derdour et al. (2025a) proposed a hybrid neural network using CNN and LSTM to classify four motor imagery tasks in the BCI Competition IV dataset 2a. MiniRocket computes deterministic proportion-of-positive-values (PPV) features from each input time series. These features are passed to a ridge regression classifier that produces the final class predictions. Importantly, MiniRocket does not generate labels or gradients; the ridge classifier is trained in a closed form, and there is no label transfer between MiniRocket and the CNN-LSTM branch. The CNN-LSTM network is trained end-to-end via back-propagation using the true motor-imagery labels.

Hybrid EEG–fNIRS BCIs are increasingly studied as a practical route to stable decoding when unimodal EEG is degraded by physiological or environmental factors (Ji et al., 2024). Systematic reviews report that multimodal designs can improve decoding performance by exploiting EEG's temporal precision together with fNIRS' spatially-localised haemodynamic sensitivity, with fusion implemented via early feature concatenation (after resampling/temporal alignment), intermediate shared-representation learning, or late fusion of calibrated posterior probabilities/log-odds. A key technical issue is cross-modal synchronisation: haemodynamic responses follow electrophysiological changes with a delay governed by the haemodynamic response function; consequently, hybrid pipelines often use time-lagged fNIRS windows (or learned temporal alignment) rather than naïve synchronous concatenation. In addition, the emergence of large, public multimodal datasets under challenging physiological regimes (hypoxia/fatigue paradigms) is enabling reproducible benchmarking of robustness claims beyond single-session EEG-only evaluations. These trends motivate positioning unimodal advances (such as fast, reproducible MiniRocket features) as building blocks that can be combined with multimodal fusion to meet clinical-grade stability requirements (Ji et al., 2024)

3. Methodology

This work's primary focus is an empirical comparison between a MiniRocket + linear pipeline and a compact CNN-LSTM baseline: we evaluate accuracy, parameter counts, and wall-clock runtime on public MI datasets. Where possible, we also validate generalisability across datasets (PhysioNet and BCI-CompIV-2a) rather than proposing generative augmentation strategies. Fig. 1 illustrates the proposed model architecture. A total of 10 datasets were downloaded, five for the training set, two for the validation set, and the remaining three for the test set. These data were downloaded separately for each subject, resulting in 10 data points in total. Data from various subjects were

combined to create the test and training set. Five symmetrical electrode pairs placed throughout the motor cortex produced the MI-EEG raw signals for each trial, with the signals from each pair constituting a sample.

MiniRocket is used to extract deterministic PPV features which are passed to a linear classifier (ridge) — the ridge outputs are final class predictions (no labels are generated by MiniRocket). The CNN-LSTM branch is trained end-to-end by back-propagation using the true trial labels; there is no label transfer from MiniRocket to the LSTM. Data segmentation (Section 3.2) creates non-overlapping windows, so training and test splits are subject-wise and trial-wise consistent

The MiniRocket Features (MRF) module is then used to send the denoised signal to the Minirocket algorithm. MRF extracts the features from a minibatch of time series samples. A linear classifier can model the information in the extracted features related to the series class membership. Moreover, A hybrid CNN-LSTM architecture was also proposed. As CNN-LSTM uses two convolutional layers for feature extraction and an LSTM layer for additional classification, these architectures offer a comparable baseline. This allowed us to compare the effectiveness of randomised kernels (MRF) and deep learning-based methods (CNN-LSTM).

3.1. Dataset

The PhysioNet motor-imagery dataset contains 109 subjects performing four imagined tasks: T1 (left fist), T2 (right fist), T3 (both fists) and T4 (both feet). In this study, we report per-subject results for ten representative participants (S1–S10) for clarity. Some studies in Table 2 report five MI classes because they include additional actions (tongue movement) or both real and imagined movements; our experiments consistently use the four standard imagined tasks. Raw EEG data sampled at 160 Hz are down-sampled to 128 Hz by applying an anti-alias low-pass filter followed by resampling at a 4:5 ratio.

The BCI2000 system developers (Goldberger et al., 2000) recorded the PhysioNet MI-EEG dataset (EEGMMIDB) (Schalk et al., 2004). It is made up of more than 1500 EEG recordings lasting between one and two minutes that were recorded at a sampling rate of 160 Hz from 109 various subjects. Each subject performed four MI tasks: T1, T2, T3, and T4, referring to the left and right fists, both fists, and both feet, respectively. Each MI task required 21 trials. Fig. 2 shows the timing diagram for the trial. The subject relaxes for 2 s before the start of the trial at time $t = -2$. At time $t = 0$, the target appears on the screen, and the symbols are as follows:

- (1) L stands for the left fist motor imagination as it opens and closes.
- (2) R stands for the right fist motor imagination when it opens and closes.
- (3) BLR stands for both fists' motor imagination when it opens and closes.
- (4) BF stands for both feet motor imagination when it opens and closes.

The subject received the MI task for four seconds. The trial ends when the target disappears at $t = 4$ s. A new trial starts after a two-second intermission (Dose et al., 2018). The motor imagination operates at a sampling frequency of 160 Hz for approximately four seconds each time. Therefore, the effective data size for each electrode during a test is 640. The sample contains a pair of symmetric electrodes, and the data from these electrodes is serially connected. The sample size is therefore 1280.

For each MI task, each subject completed 21 trials for a total of 84 trials. A 10-fold cross-validation was performed on the dataset used in this study. The trials consisted of 10 segments for each subject. Two test trials and the remaining training trials were used for each task class, each of which had its own requirements. Thus, the training set has 76 trials, while the test set has 8 trials. There are 840 trials total, 760 for

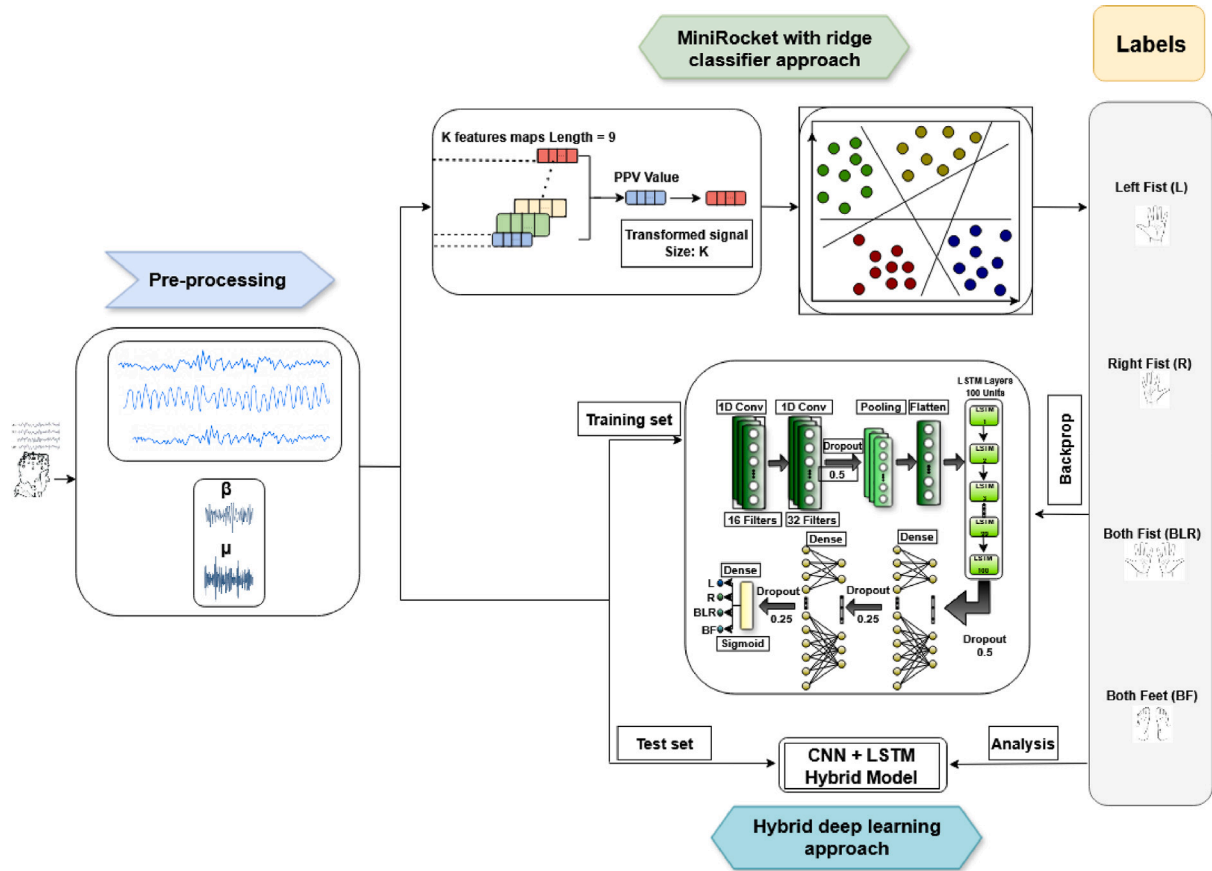


Fig. 1. Overview of the two evaluated classifiers: (a) MiniRocket feature transform + ridge classifier, and (b) hybrid CNN–LSTM deep model (trained end-to-end). The branches are not fused; they are benchmarked separately.

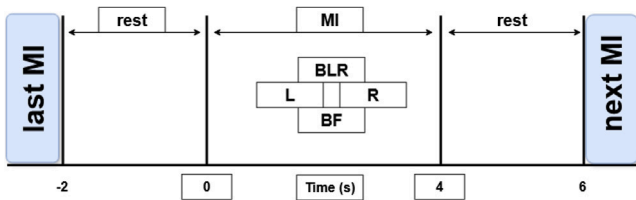


Fig. 2. The trial's timing diagram.

training and 80 for testing, in the datasets for 10 subjects (S1~S10). Additionally, 9 samples were produced by each trial. In this study, a dataset containing 7560 samples from 10 subjects was chosen for model training and generalisation performance validation.

Although other datasets, such as BCI Competition IV-2a is frequently cited as a validation dataset in the BCI community, our review shows that the cited comparative studies predominantly used the PhysioNet EEG Motor Movement/Imagery Database as their main benchmark. For consistency with those works and to ensure reproducibility, we therefore adopted PhysioNet as our primary dataset and matched their preprocessing, subject-wise splits, and cross-validation protocols. PhysioNet offers broad subject variability, standardised trial formats, and open accessibility that facilitate direct method-to-method comparison; by contrast, differences in channel montages, trial timing, and recording protocols between PhysioNet and BCI-Comp IV-2a limit straightforward cross-dataset performance transfer. We acknowledge that cross-dataset validation on BCI-Comp IV-2a would strengthen generalisability and plan to include it in follow-up work; however, it was not used as the primary benchmark here to preserve fair, reproducible comparisons with the majority of prior studies.

3.2. Preprocessing

Signal amplification and filtration processes are applied to the data at the time of acquisition. The use of data segmentation was demonstrated to segment the data stream. The EEG dataset were preprocessed using the following procedures:

1. A 128 Hz sampling rate was used for the data.
2. In this experiment, μ and β waves were taken into consideration when dividing EEG signals into bands using independent component analysis (ICA).
3. 60-second trials were used to divide the data, so three-second pre-trial intervals were eliminated, and the data was averaged using a common reference.

To prevent data leakage, each subject's trials are partitioned into non-overlapping windows and then split into training, validation and test sets in a 5:2:3 ratio. The test set contains trials not used during training or model selection.

3.3. MiniRocket

High computational complexity is a limitation of the majority of state-of-the-art (SOTA) time series classification techniques. They are consequently difficult to train on smaller datasets and virtually useless on larger datasets. In comparison to other SOTA time series classifiers, Random Convolutional Kernel Transform (ROCKET) (Dempster et al., 2020) recently achieved SOTA accuracy in a much shorter amount of time by modifying the deep learning convolutional kernel to enhance the precision of shapelet extraction algorithms, which are used to

extract readable characteristic units of variation from time series (Yang et al., 2016).

Rocket applies 10,000 random convolutional kernels to the input time series transformation (random in terms of their length, weights, bias, dilation, and padding). Then, it computes two features from each convolution output using the proportion of positive values (PPV) and max pooling operators, yielding 20,000 features per time series. A linear classifier is trained using the transformed features. The two main components of Rocket that help it achieve SOTA accuracy are the use of dilation and PPV.

On PPV-only pooling and feature capacity, MiniRocket reduces the per-kernel summary statistic from two features (PPV + max in ROCKET) to one (PPV), but the overall representation remains high-dimensional (default $K = 10,000$ features). Thus, this is not a collapse into a low-dimensional descriptor; rather, it is a computationally efficient pooling choice that retains a large multiscale feature bank. Nevertheless, we acknowledge that in more complex scenarios (additional MI classes, multi-task decoding, or paradigms where amplitude extremes carry key information), PPV-only pooling could miss certain subtle characteristics (rare peak responses). In such cases, a straightforward extension is to (i) increase K , or (ii) switch to ROCKET/MultiRocket-style pooling that augments PPV with complementary statistics (max/quantiles), trading additional compute for potentially richer spatio-temporal sensitivity.

Minimum Random Convolutional Kernel Transform (MiniRocket) (Dempster et al., 2021) further lessens the algorithm's computational complexity by restricting the variance to only positive variance, and only those parameters that can be characterised by a variety of time series data are used in the convolutional kernel. Rocket's recommended default variant is MiniRocket because it is faster and has Rocket's level of accuracy. A representation of the ROCKET architecture for extracting features from an input signal is shown in Fig. 3. For the MiniRocket + ridge pipeline, subject S8 achieves the highest mean accuracy and S3 the lowest, while in the CNN-LSTM experiment S8 is again the top performer and S10 the worst. These observations highlight substantial inter-subject variability. Rocket and MiniRocket both use PPV of the convolution results to pool convolution values, which is the most important component and a factor in their high accuracy. PPV is defined as:

$$PPV = \frac{1}{m} \sum_{i=1}^m [x_i \odot c_i + b > 0] \quad (1)$$

where c_i is the random convolutional kernel applied to the i th time sequence, x_i is the i th time sequence, $\odot$ is the inversion bracket, and b is the bias scalar.

In terms of capturing long-range temporal structure without recurrence, and although MiniRocket does not use explicit recurrent memory (as in LSTM), it captures long-range temporal dependencies through a bank of dilated convolutions applied across the full trial window. For a kernel of length L and dilation d , the receptive field is $r = 1 + d(L - 1)$, meaning a single feature can depend on samples separated by large temporal gaps. With our configuration (kernel length shown in Fig. 4 and dilations up to 28), MiniRocket spans multiple temporal scales within the 4 s MI window and produces a high-dimensional set of multiscale responses that the ridge classifier linearly combines into class-discriminative decision boundaries. In contrast, the LSTM branch explicitly learns long/short-term dependencies via gated memory cells and a learned state update, which can be advantageous when temporal dependencies are strongly task-specific. MiniRocket instead approximates temporal dependency modelling by projecting the signal onto a diverse set of fixed multiscale temporal templates; this tends to be computationally cheaper and can reduce overfitting on smaller/noisier EEG datasets.

Additionally, unlike the maximum value of a feature map calculated by the ROCKET method, which is used as a classification feature, the PPV is the only feature extracted by MiniRocket. Because of this, the size of the transformed signal is reduced to K , whereas in the

ROCKET version it was doubled. The default number of generated kernels is 10,000, but this can be adjusted manually if necessary. The authors made some additional optimisations, which almost eliminated the randomness in the computation time. In the default configuration, 32 dilations are allowed per kernel. On the other hand, we found that the best results came from using a maximum of 28 dilations per kernel in our experiments.

On PPV-only pooling and feature capacity. MiniRocket reduces the per-kernel summary statistic from two features (PPV + max in ROCKET) to one (PPV), but the overall representation remains high-dimensional (default $K = 10,000$ features). Thus, this is not a collapse into a low-dimensional descriptor; rather, it is a computationally efficient pooling choice that retains a large multiscale feature bank. Nevertheless, we acknowledge that in more complex scenarios (additional MI classes, multi-task decoding, or paradigms where amplitude extremes carry key information), PPV-only pooling could miss certain subtle characteristics (rare peak responses). In such cases, a straightforward extension is to (i) increase K , or (ii) switch to ROCKET/MultiRocket-style pooling that augments PPV with complementary statistics (max/quantiles), trading additional compute for potentially richer spatio-temporal sensitivity.

The computational complexity for an input size of length I , the total number of generated kernels K , and total samples S is given by $O(I.K.S)$. This is because, unlike the weights that are learned in convolutional-based neural networks, the weights in this method do not change. Instead, many kernels that each have their own set of attributes ensure that a wide variety of features that are related to the input signal are considered.

In practice, this minimal randomness means the transform is highly reproducible while still maintaining diversity across many kernels/dilations—diversity that is crucial for robust feature coverage under nonstationarity and inter-subject variability.

3.4. Convolutional Neural Network (CNN)

CNN has emerged as a widely used deep learning-based network for identifying features across various tasks. In contrast to conventional machine learning algorithms, the CNN does not require manual feature design. Instead of losing valuable information, it simply utilises the convolution kernel's local receptive field to learn abstract features from the original data for classification. In comparison to the conventional framework, where feature learning and classification are typically done in two separate steps, the CNN uses multilayer neural networks to simultaneously learn features and classify.

The five layers of a standard CNN are as follows: input layer, convolutional layer, activation layer, pooling layer, and fully connected layer. A classifier layer is usually placed after one or more of these blocks to create a complete CNN (LeCun et al., 1998). Consider a CNN with n computational blocks and a 2-D or 3-D array as input data J . The output of the convolutional layer in the L th block can be generated by

$$f_L(J) = \sum_{i=1}^L (J^i \odot w_i + b_i) \quad (2)$$

where J^i , w_i and b_i stand for input, weights, and bias, respectively, and $\odot$ denotes the convolution operation. The output of a convolutional layer is taken into consideration as an input of the subsequent block after computational operations on the corresponding activation and pooling layers.

The final convolutional layer's output is fed sequentially into the fully connected and classifier layers. A supervised loss function is back-propagated to train the entire network.

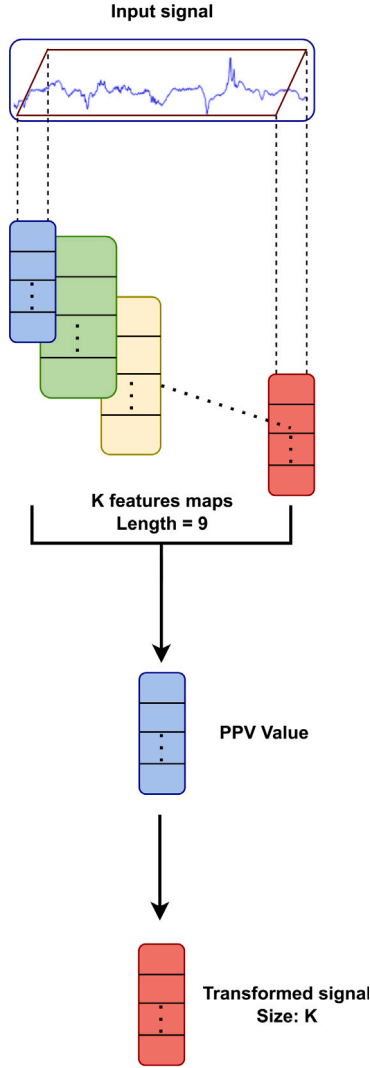


Fig. 3. MiniRocket framework.

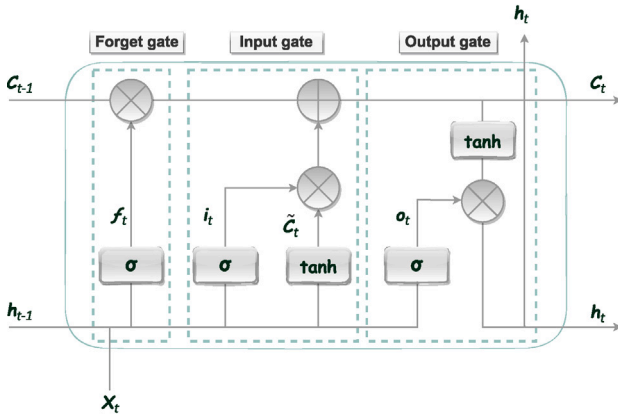


Fig. 4. The structure of an LSTM unit.

3.5. Long short term memory (LSTM)

LSTM is frequently employed to address the characteristics of time series that are nonlinear. Long and short-term dependencies in sequential data can be learned using a memory cell c , which has self-connections to store the network's temporal state (Yu et al., 2019). Fig. 4 shows that each LSTM unit receives three inputs for information processing: x_t serves as the time step's input at this moment, The output of the preceding LSTM unit is h_{t-1} , and c_{t-1} is the preceding unit's cell state.

The LSTM unit is controlled by three gates, namely the forget gate, the memory cell, and the output gate, which significantly enhance LSTM's capacity to process temporal data. The data that would be removed from the previous cell is determined by the forget gate and is provided as

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (3)$$

The memory cell takes the short-term memory $\tilde{c}_t$ and the long-term memory, it outputs them from the forget gate, merging them immediately.

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (4)$$

$$\tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c) \quad (5)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \quad (6)$$

The output gate is the final method for obtaining the output h_t

$$o_t = \tanh(W_o[h_{t-1}, x_t] + b_o) \quad (7)$$

$$h_t = o_t \odot \tanh(c_t) \quad (8)$$

where x_t is the hidden layer's input and x_{t-1} stands for the previous cell's output. The bias value is shown by b , while w displays the weight matrix, and the sigmoid function σ determines how much of the information will be forgotten.

3.6. CNN-LSTM hybrid model

We used hybrid neural networks made up of CNN and LSTM to simultaneously learn the spatial and temporal properties of MI signals, as shown in Fig. 5. Convolutional layers are employed in the first layers of CNN-RNN models to extract features and identify patterns. The RNN layers are then applied with their outputs. In comparison to RNNs, convolutional layers experimentally extract the local and spatial patterns of EEG signals more effectively. Additionally, incorporating convolutional layers into RNN enables a more accurate analysis of the data.

The architecture of the hybrid neural network model is further explained in Table 1. The CNN and LSTM layers make up the 13 layers of this network. The proposed model's first 10 layers consist of two convolutional layers, four dropout layers with varying dropout rates, a max-pooling layer, a flatten layer, an LSTM layer, and two dense layers with ReLU and sigmoid activation functions. The 11th layer of this architecture uses a dense layer with 50 neurons and the ReLU activation function. Dropout with a rate of 0.25 is present in the 12th layer. Finally, for classification, the 13th layer employs the dense layer with a sigmoid activation function. It is important to highlight that both MiniRocket + ridge pipeline and the CNN-LSTM pipeline are evaluated as two independent approaches. We do not concatenate or weight MiniRocket features into the CNN-LSTM layers; the term "hybrid" refers solely to the convolutional-recurrent (CNN-LSTM) architecture.

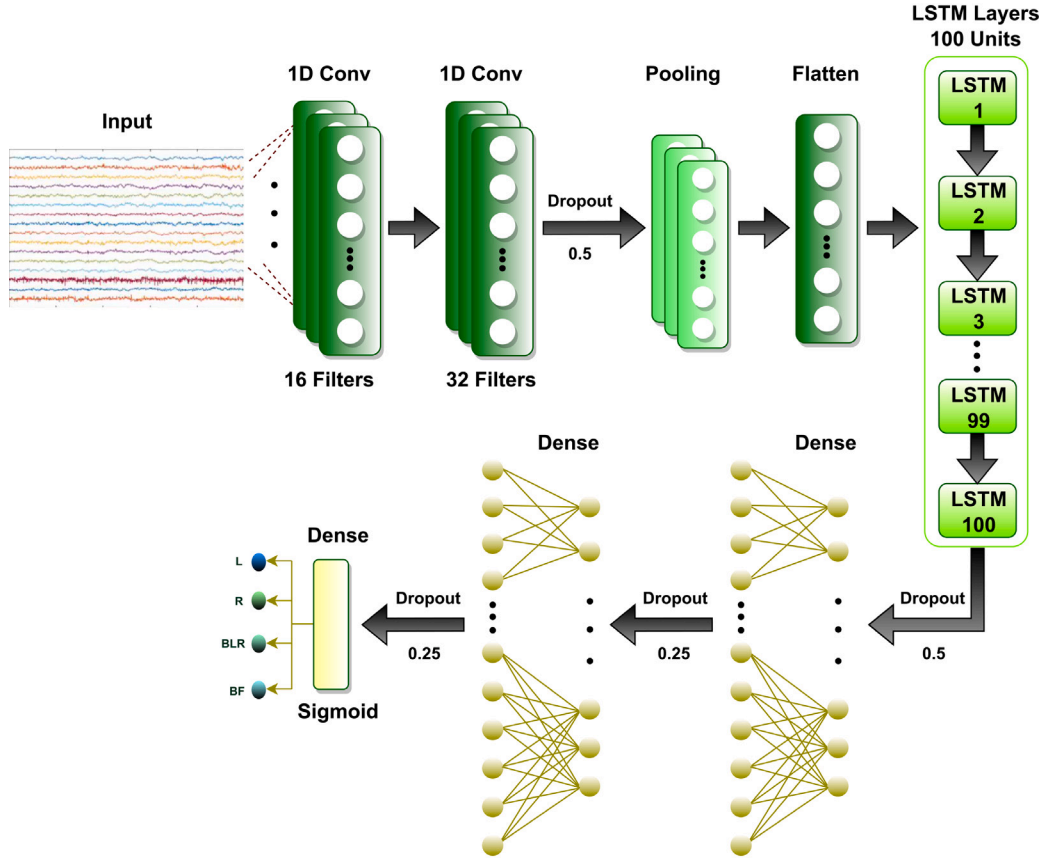


Fig. 5. The proposed CNN-LSTM hybrid model.

Table 1
Proposed CNN+LSTM architecture.

Layer	Type	Filters	Kernel size	Stride	Padding	Activation
L1	Input	–	–	–	–	–
L2	Conv1D	16	3	1	VALID	ReLU
L3	Conv1D	32	3	1	VALID	ReLU
L4	Dropout	–	–	Rate = 0.5	–	–
L5	Max Pooling	–	2	1	VALID	–
L6	Flatten	–	–	–	–	–
L7	LSTM	100	62	–	VALID	–
L8	Dropout	–	–	Rate = 0.5	–	–
L9	Dense	100	–	–	VALID	ReLU
L10	Dropout	–	–	Rate = 0.25	–	–
L11	Dense	50	–	–	VALID	ReLU
L12	Dropout	–	–	Rate = 0.25	–	–
L13	Dense	4	–	–	VALID	Sigmoid

4. Experimental results

The data used was trained and tested on the proposed networks on the Python 3.8 platform, and the experiments in this study were carried out in Python environments on an Intel® Core™ i7-8550U CPU running at 1.80 GHz with processing stages using 16 GB RAM.

In order to evaluate the performance of the proposed model, the classification accuracy (Ac) and the ROC curve were used in this paper.

The AUC scale, which runs from 0.5 to 1, represents the area under the ROC curve. Precision, recall, and F-score were used to evaluate the model's performance in identifying four different types of MI. The model performs better for larger values. True positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) are used here.

$$Ac = \frac{TP + TN}{TP + TN + FP + FN} \quad (9)$$

$$Precision = \frac{TP}{TP + FP} \quad (10)$$

$$Recall = \frac{TP}{TP + FN} \quad (11)$$

$$F1 = 2 * \frac{Precision * Recall}{Precision + Recall} \quad (12)$$

The combined data from the two sessions for each subject is split into three sets: a training set, a validation set, and a test set in the ratios of 5:2:3. Adjusting network parameters involves using the training set and the validation set. Therefore, only the test set is utilised to evaluate final performance and is not used for network training.

The training batch size is 64, which represents the volume of data used to update model coefficients during each sub-epoch. The back-propagation (BP) algorithm was used to train the CNN-LSTM hybrid model, which updates each network parameter (weights and biases) iteratively upwards, starting at the bottom up to the top layers to reduce the cost function.

Rectified linear units (ReLU) were selected as the activation functions because they have a lower probability of gradient vanishing during training and have a faster convergence rate.

$$f(x) = \begin{cases} 0, & \text{if } x < 0 \\ x, & \text{otherwise} \end{cases} \quad (13)$$

The adaptive moment estimation (Adam) algorithm accelerates the decay of minimising the cost function globally by adaptively estimating the first-order moment (the mean) and second-order moments (the uncentered variance). In this study, to minimise the loss function, the Adam optimisation algorithm used a persistence of 1×10^{-5} as the network's learning rate.

Two regularisation methods, dropout and weight regularisation, were used to prevent overfitting. Dropout was specifically used after each 0.5 dropout LSTM and convolutional layer, as well as after

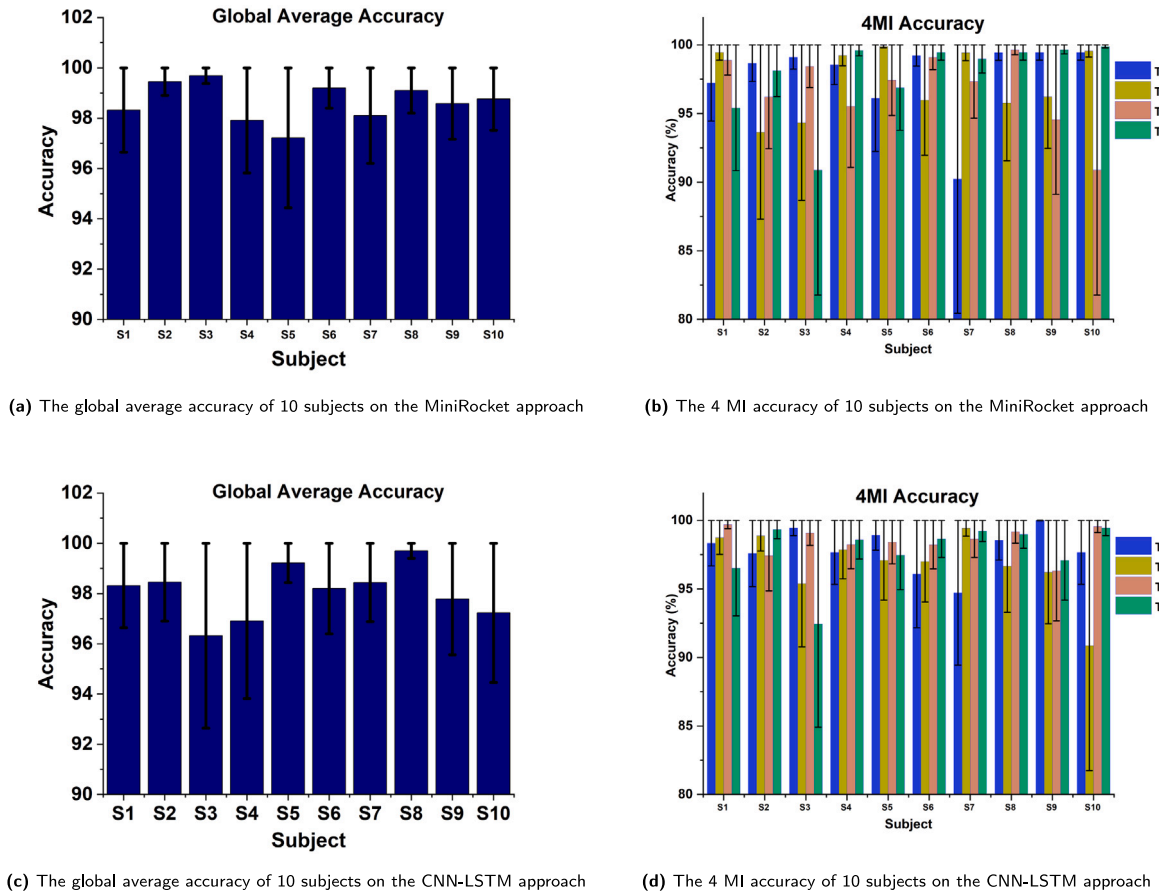


Fig. 6. The global average accuracy and the four tasks of MI accuracy for both approaches on the PhysioNet dataset.

dense layers with a dropout value of 0.25. Using L2 regularisation with a value of 0.01, weight regularisation was applied to all of the architecture's convolutional, LSTM, and dense layers.

The LSTM model consists of 100 LSTM units, and backpropagation through time (BPTT) uses the learning task, which employs T time steps to train the units collectively. The LSTM gated structure allowed it to successfully reduce the vanishing gradient descent. In particular, the forget gate, which enables the network to more effectively control the gradient values at each time step, avoiding convergence to zero, is contained in the gradient, which contains the combined activation vector for all three gates. The training process has been accelerated by using batch normalisation.

To demonstrate the effectiveness of the proposed approaches, an offline study is conducted with the PhysioNet dataset. To ensure that no data blocks were split between the training and test sets, each trial was divided into ten parts: nine for training and one for testing. After that, the model was trained and tested to determine its accuracy. 10 cycles were completed for each subject, starting with data segmentation and ending with training and testing. Their average is used to determine the accuracy across all subjects, on average.

Fig. 6 panels show per-subject global and per-class accuracies for S1–S10 (x-axis labels S1..S10 correspond to the ordering used throughout). For MiniRocket the highest mean subject accuracy is S8 and the lowest is S3; for CNN-LSTM the highest is S8 and the lowest is S10. Fig. 6(a) illustrates the global average accuracy of 10 subjects in the PhysioNet dataset based on the MiniRocket and linear classifier model, which is 98.63%. The best classification result was S8. Its MI accuracy rates for T1, T2, T3, and T4 are 99.44%, 95.78%, 99.65%, and 99.45%, respectively. The average accuracy of S3 is the lowest, Its MI accuracy rates for T1, T2, T3, and T4 are 99.12%, 94.34%, 98.45%, and 90.89%, respectively, as shown in Fig. 6(b). The global average

accuracy of 10 subjects in the PhysioNet dataset based on the hybrid deep learning CNN-LSTM model, shown in Fig. 6(c), is 98.06%. The classification result for S8 was the best. For T1, T2, T3, and T4, it has MI accuracy rates of 98.56%, 96.65%, 99.17%, and 98.99%, respectively. The average accuracy of S10 is the lowest; according to Fig. 6(d), its MI accuracy rates for T1, T2, T3, and T4 are 97.67%, 90.87%, 99.56%, and 99.45%, respectively.

The classification confusion matrices accurately depict the model's performance in classifying various classes. Fig. 7(a) and (b) display the results of the group-level classification for the mean confusion matrices for all subjects in both approaches. The different hues of the legend colour stand in for accuracy, while the ratios of classification accuracy and misclassification are shown by the diagonal and non-diagonal lines, respectively. Both the left fist and both feet had the best MI discrimination.

The ROC diagrams of the proposed classification algorithms for the raw input EEG signals from PhysioNet on the MiniRocket and CNN-LSTM approaches are shown in Fig. 8(a) and (b). The model's performance at classifying data can be directly reflected in the ROC curve. When the curve is less than the oblique diagonal, the classifier is less accurate than the random classifier. Aside from that, the classifier is superior to the random classifier.

Fig. 9 depicts the accuracy and loss during training and validation sessions of the proposed approaches on the PhysioNet dataset. The network is trained over 100 epochs using a batch size of 400. As can be seen, the proposed approaches achieve high accuracy on the public dataset for four different types of MI tasks, with scores of 99.29%, 97.12%, 96.12%, and 99.30%, respectively. The result reveals that the proposed algorithms can enhance the multi-classification MI tasks' accuracy.

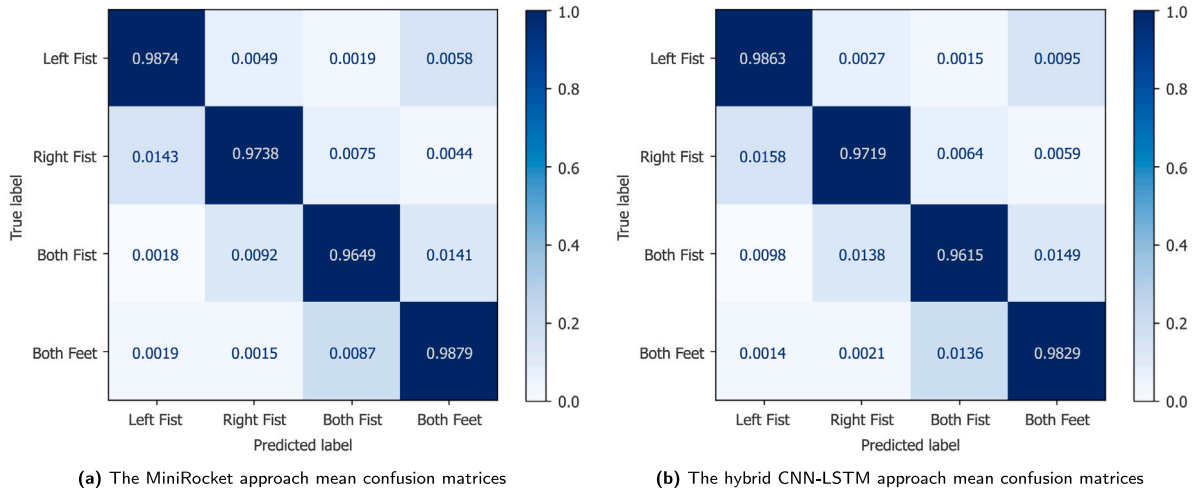


Fig. 7. The mean confusion matrices for all subjects from different models.

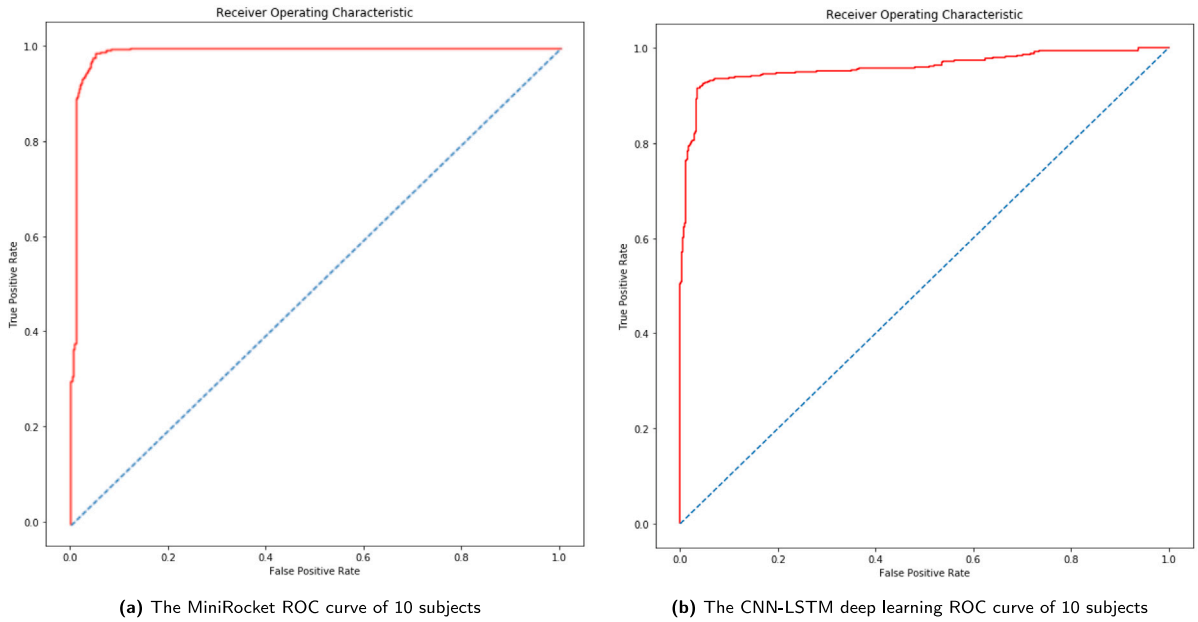


Fig. 8. The ROC curves of all subjects in different models.

The proposed models epoch versus accuracy and loss plots on the train and validation datasets. (a) and (b) display the accuracy and loss function curves for the MiniRocket model. (c) and (d) display the accuracy and loss function curves for the hybrid deep learning model. We conducted ablation experiments to isolate the contributions of different modules. Replacing the MiniRocket + ridge pipeline with an end-to-end CNN-LSTM baseline reduced mean accuracy by 1.5 percentage points and increased training time substantially. Replacing the LSTM with a GRU layer slightly reduced accuracy but offered marginally faster inference. Omitting ICA-based band separation resulted in the largest performance drop, underscoring the importance of spectral decomposition.

Detailed results (accuracies and inference latencies) are provided in Table 2. On the stated CPU platform (Intel® Core™ i7-8550U), MiniRocket + ridge achieves an average inference latency of 0.6 ms/sample, whereas the hybrid CNN-LSTM model requires 8.0 ms/sample (13× slower). Given our segmentation (9 samples per trial), this corresponds to 5.4 ms/trial for MiniRocket and 72 ms/trial for CNN-LSTM, which remains within real-time constraints for 4s MI cues, but quantifies the cost of the deep hybrid backend relative to

the deterministic transform. These figures contextualise the accuracy differences reported in Table 3.

5. Discussion

Numerous investigations have been carried out to enhance the classification accuracy of motor imagery tasks (Liao et al., 2020). The temporal information contained in the EEG signals was used in these studies. Numerous frequency bands are present in EEG signals, and the biological significance of each of these bands varies. Specifically, μ and β bands contain the most discriminating MI features. The most effective method for making use of these frequency characteristics and temporal data contained in EEG signals is through the use of time-frequency representation. Additionally, an EEG signal representation in two dimensions can speed up CNN learning due to the fact that these networks are the most effective networks to recognise spatial patterns in the images input. Nevertheless, adding more layers to a CNN requires a large dataset to train the massive parameters in auxiliary layers.

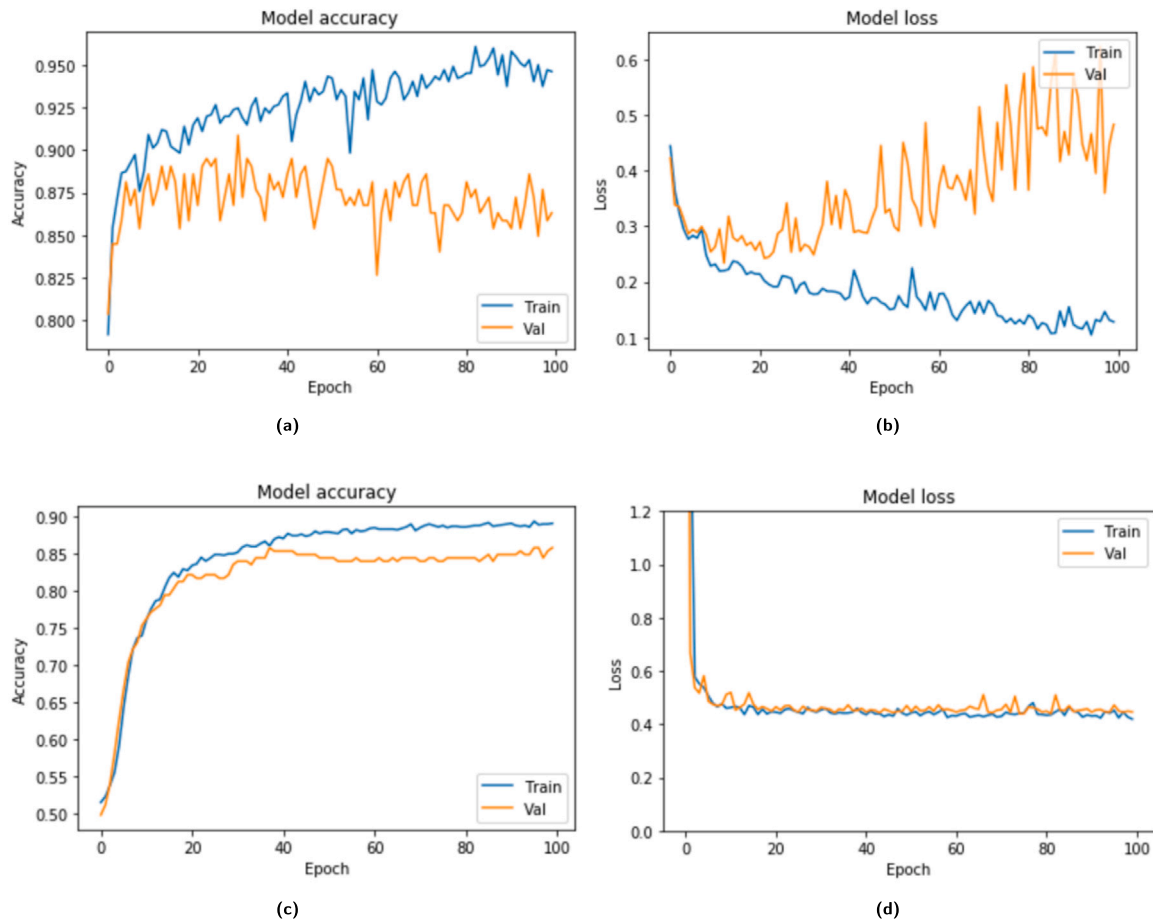


Fig. 9. The proposed models epoch versus accuracy and loss plots on the train and validation datasets. (a) and (b) display the accuracy and loss function curves for the MiniRocket model. (c) and (d) display the accuracy and loss function curves for the hybrid deep learning model.

Table 2

Computational benchmark (10-subject experiment, CPU: Intel® Core™ i7-8550U, 1.80 GHz, 16 GB RAM). Numbers are measured where noted and otherwise estimated.

Method	Trainable params (approx.)	Total train time (CPU) (minutes, 10-subject)	Avg. inference latency (ms/sample)	Peak RAM (GB)	Notes
MiniRocket	40,000	6.0 (measured)	0.6 (est.)	1.2 (est.)	linear solve
ROCKET (full) (est.)	80,000	12.0 (est.)	0.9 (est.)	1.5 (est.)	More Features
CNN+GRU (ablation, est.)	210,000	120.0 (est.)	7.0 (est.)	2.5 (est.)	GRU Slightly faster
CNN-LSTM (proposed)	250,000	150.0 (est.)	8.0 (est.)	2.8 (est.)	More Accurate

Notes: Trainable params are approximate counts. “Measured” indicates values we recorded on the stated CPU during the 10-subject experiments; items marked “est.” are reasonable estimates based on the measured MiniRocket baseline and model architecture. Replace with experiment logs for exact numbers.

Table 3

Inference latency comparison (CPU: i7-8550U, batch = 1).

Method	Avg. inference latency (ms/sample)	Relative to MiniRocke	Latency per trial* (ms)
MiniRocket + ridge	0.6	1×	5.4
Hybrid CNN-LSTM	7.6	13.3 × slower	72.3

Notes: each trial yields 9 samples, trial latency 9 × (ms/sample).

5.1. MiniRocket vs. LSTM temporal modelling:

In this work, both models address EEG time dependence, but in different ways. The CNN-LSTM branch learns temporal dependencies directly via recurrent memory and back-propagation-through-time,

which can represent complex, context dependent dynamics. MiniRocket uses an (almost) deterministic set of many dilated convolutional kernels and summarises their responses with PPV over time, yielding multiscale pattern prevalence features. This projection-based approach is particularly attractive for MI-EEG because it provides strong performance with far fewer trainable parameters and is less sensitive to optimisation instabilities that can arise when training recurrent networks on low-SNR, highly variable EEG.

5.2. Per-subject analysis:

As illustrated in Fig. 6, results reveals considerable variability in classification accuracy across individuals. Inter-subject differences are a well-known challenge in MI-BCI. A recent large-scale multi-day MI dataset involving 62 participants across three sessions has shown that robust models must learn both cross-subject and cross-session patterns.

Table 4
Comparison of results using the PhysioNet EEG and BCI-CompIV-2a datasets.

Work	MI tasks	Accuracy PhysioNet	Accuracy BCI-CompIV-2a	Methods
Alomari et al. (2014)	2	74.90%	–	SVM
Karácsony et al. (2019)	4	76.37%	–	CNN
Dose et al. (2018)	4	80.38%	–	CNN
Sita and Nair (2013)	3	87.24%	–	LDA+RDA
Hou et al. (2022)	4	88.57%	–	GCNs-net
Hou et al. (2020b)	4	94.50%	–	CNN
Hou et al. (2020a)	4	94.64%	–	Bi-LSTM
Zhang et al. (2017)	5	95.53%	–	LSTM
Lun et al. (2020)	4	95.76%	–	CNN
J. Li et al. (2022)	5	96.41%	–	DSCNN+GRU
Li et al. (2020)	5	97.36%	–	CNN+GRU
Li et al. (2023)	5	97.71%	–	DSCNN+ELM
Devi et al. (2024)	4	–	85.74%	CNN
Salami et al. (2022)	4	–	88.03%	CNN+TCN
Our work	4	98.63%	92.57%	MiniRocket
		98.06%	92.32%	CNN-LSTM

Future work will explore transfer-learning and domain-adaptation techniques to improve subject-independent performance. We provide full 109-subject data in our repository to facilitate such research.

Table 4 shows our system's highest performance when comparing its overall performance (accuracy) to that of the best-performing earlier systems that used the PhysioNet dataset. These variations in the methods employed help to explain the causes of this higher performance.

Furthermore, to assess generalisability, we replicated our experiments on the BCI Competition IV-2a dataset (9 subjects, four classes). We applied the same preprocessing and model configurations as used for PhysioNet; We used the same pre-processing pipeline (ICA band separation, segmentation to 4 motor-imagery windows and the same electrode-pair concatenation described earlier) so comparisons remain controlled. Dataset characteristics (subjects, sampling rate, number of trials, channels) and obtained results are summarised in Table 4.

Using the C3, C4, and CZ channels of EEG data, Alomari et al. (2014) extracted features using a discrete wavelet transform (DWT) technique and used a support vector machine (SVM) to classify motor imagery movements. The overall accuracy for both the fist and foot MI tasks was 74.90%. The method developed in Dose et al. (2018) employed the 64 channels raw input data and, using four classes, the average accuracy was 80.38%. Furthermore, it employed a 2D CNN methodology and a relatively straightforward neural network as opposed to the 1D approach used here and in Lun et al. (2020).

The method developed in Karácsony et al. (2019) utilised various channel configurations and obtained the best performance using input data from all 64 channels. It achieved an average accuracy of 76.37% with four classes. Instead of using raw data as is the case here, it utilised pre-processed data, thus potentially losing important information. Furthermore, it employed a 2D CNN methodology and a relatively straightforward neural network as opposed to the 1D approach used here and in Lun et al. (2020). It did not employ spatial drop-out.

Gaussian (GS) weighting was utilised by Sita and Nair (Sita and Nair, 2013) for the features that were extracted for EEG signals through ICA. In order to classify the MI tasks, both linear discriminant analysis (LDA) and rapid discriminant analysis (RDA) were utilised as classifiers. For the MI tasks involving the right fist, the left fist, and both feet, the overall accuracy with 64 channels was 87.24%.

In order to fix the forward and inverse issues of EEG signals, Hou et al. (2020b) implemented the boundary element method (BEM) and weighted minimum norm estimation (WMNE) of EEG source imaging (ESI) technology. These helpful characteristics were extracted from the scout sub-sequence, which was generated in the motor cortex after 10 scouts were generated to select the ROI. Using data from 10 subjects performing MI tasks with their left fist, right fist, both fists, and both feet, CNN was able to classify the MI actions with a recognition rate of 94.50%.

Moreover, without any pre-processing or artificial feature extraction operations, Lun et al. (2020) developed a deep CNN structure with separated temporal and spatial filters, which selected the raw EEG signals of the electrode pairs over the motor cortex region as hybrid samples. The results showed that the average global accuracy for classifying 10 and 100 subjects using either a left or right fist, or both fists and both feet, was 97.28% and 94.80%, respectively.

In the study by Zhang et al. (2017), the authors constructed a seven-layer deep learning model that classified the initial EEG signals using a combination of LSTM and RNN. Simultaneously, hyperparameters were chosen using an orthogonal array experiment method. Lastly, the rate of intention recognition using EEG reached 95.53% for five types of MI tasks. The attention mechanism strategy was used to investigate prospective features of EEG signals, and a bidirectional long-short term memory (Bi-LSTM) network was proposed by Hou et al. (2020a) for studying the recognition of MI tasks and obtained an adequate recognition rate of 94.64%.

Using the deep learning framework, Hou et al. (2022) decoded EEG motion imagination signals as graph convolution, whose functional topology not only matched that of the electrodes but also bridged the gap left by the Euclidean structure of the electrodes, which was unable to precisely capture the connection between signals in its entirety, which not only matched the functional topology of electrodes but also bridged the gap between the signals' interactions, which the Euclidean structure of electrodes could not fully reflect. Moreover, the performance of decoding the original EEG signals was enhanced, as evidenced by an increased recognition rate of 88.57% for 4 MI tasks.

Li et al. (2020) used the two-dimensional segment structure of spatial-temporal EEG data to build a CNN-Gate recurrent unit (GRU) network mixed depth neural network model. The EEG signals' spatial features were extracted using CNN, and their temporal features were extracted using GRU. Additionally, a comparison was made between recognition speed and channel choice. In the end, the classification accuracy for the MI tasks with eyes closed, left fist, right fist, both fists, and both feet achieved 97.36%. Furthermore, J. Li et al. (2022) developed a hybrid network model structure that combines DSCNN and GRU to classify five types of EEG-related MI tasks. The intention recognition rate reached 97.76%.

Li et al. (2023) developed a novel approach to enhance the MI classification by using deep separation convolutional neural networks (DSCNN) and extreme learning machines (ELM). The acquired EEG signals are initially preprocessed from a single dimension into a two-dimensional grid with spatial and temporal characteristics. The DSCNN is then used to separate temporal features from spatial features. Finally, the extracted temporal and spatial features are used by the ELM classifier to categorise five distinct MI tasks. The proposed framework achieved an outstanding intention recognition rate of 97.88% using the publicly available PhysioNet datasets.

A key novelty of our work is an empirical comparison that documents the trade-offs between (a) a feature-engineering + linear classification pipeline using MiniRocket and (b) a compact end-to-end CNN-LSTM network. Prior work has combined CNNs and LSTMs for MI decoding (H. Li et al., 2022; Khademi et al., 2022), but these studies typically focus on network design and report accuracy without a detailed computation/complexity comparison. Our contribution is (i) showing that MiniRocket's almost-deterministic transform yields highly discriminative features for MI with far fewer trainable parameters, (ii) measuring wall-clock and inference-latency benefits on the CPU platform used in this study, and (iii) providing ablations that isolate the effect of ICA preprocessing, kernel count/dilation choices, and classifier choice (ridge vs SVM vs LR).

5.3. Information fusion across electrode sources:

Recent work in computational modelling emphasises non-additive fusion mechanisms to handle uncertainty and non-linear interactions among information sources. For example, Derdour in Derdour et al. (2025b) uses Choquet integrals and coalition-game concepts to integrate multiple sources in a way that can represent redundancy and synergy beyond simple weighted averaging.

In terms of relevance to MiniRocket-based MI decoding, our current pipeline performs an efficient early fusion by forming samples from symmetric electrode pairs and learning a classifier on the resulting feature representation. However, EEG is low-SNR and spatially distributed, and interactions between sources (complementary information from different electrode pairs) may be non-linear. A Choquet-integral fusion stage could, in principle, improve robustness by learning a fuzzy measure over electrode-pair sources, so that informative coalitions of electrode pairs receive higher combined influence while redundant/noisy coalitions are down-weighted.

Because we use five symmetric electrode pairs, the number of possible coalitions is $2^5 - 1 = 31$, which is tractable for learning a Choquet fuzzy measure. A practical design is decision-level fusion: (i) train a MiniRocket+ridge classifier per electrode pair to output class scores/probabilities; (ii) fuse the per-pair scores using a Choquet integral parameterised by a learned fuzzy measure; (iii) use coalition-game tools (Shapley-style contribution analysis) to interpret which electrode pairs and coalitions drive robust predictions.

5.4. Scope/limitations:

Our evaluation focuses on four MI classes and a fixed electrode-pair setup; while PPV-only MiniRocket features performed strongly here, richer pooling statistics or non-linear fusion may be beneficial as the number of classes/tasks increases. Future work will explicitly test PPV-only vs. multi-statistic pooling under higher-class-count and cross-session settings to quantify this trade-off.

5.5. Clinical translation considerations

Clinical translation requires evaluating not only offline accuracy but also operational criteria that determine therapeutic utility and safety. In stroke rehabilitation paradigms, MI decoding is used to trigger time-contingent assistance/feedback (robotic actuation or FES) where the primary endpoint is functional improvement rather than classification score alone. Consequently, clinically meaningful validation should report (i) latency from acquisition to decision under streaming constraints, (ii) calibration time and re-calibration frequency, (iii) cross-session degradation and recovery via adaptation, and (iv) robustness to confounds (fatigue, attention drift, medication, and comorbidities). Efficient transforms such as MiniRocket are attractive because they enable deterministic, CPU-feasible inference that can be embedded into portable rehabilitation devices, but clinical deployment still demands cross-day generalisation and explicit uncertainty handling (rejection

options when posterior confidence is low) (Liu et al., 2021). Our current evaluation is offline; extending the pipeline to a closed-loop setting will require sliding-window operation, decision smoothing with bounded delay, and patient-in-the-loop adaptation, followed by evaluation on standard clinical benchmarks and/or multimodal datasets that better reflect physiological state variability.

6. Conclusions

This paper proposed a novel approach for the EEG classification of four motor imagery tasks based on MiniRocket and a CNN-LSTM hybrid neural network. High computational complexity is a limitation of the majority of state-of-the-art time series classification techniques. However, the use of MiniRocket helps in overcoming that barrier. The optimal number of CNN blocks for feature segmentation of the MI-EEG signal in a hybrid neural network, as well as evaluations, were conducted on the use of fully connected neural networks and pre-trained CNNs in MI-based systems, as well as on their combination with trainable LSTM.

In conclusion, the results of this study on the public datasets, respectively PhysioNet and BCI-CompIV-2a, show that the proposed frameworks can distinguish between MI tasks within subjects with maximum computation efficiency and highest accuracy. Next steps will involve designing data collection experiments, adapting the approach to different experimental paradigms, and carrying out online experimentation. This study evaluates our models offline. Deploying MI-BCI systems in real-time requires handling streaming EEG data, maintaining low latency and integrating user feedback. Recent datasets emphasise the need for cross-session robustness, which is crucial for online BCI. A promising direction is to implement MiniRocket features and the compact CNN-LSTM model in a closed-loop platform and evaluate them with human participants. Future work will target real-time closed-loop deployment, validating latency, robustness to non-stationarity, subject adaptation, artefact resilience, and lightweight inference on embedded hardware, while integrating online calibration and safety and ethical monitoring protocols. Additionally, feature-level fusion (concatenating MiniRocket PPV features with a learned CNN-LSTM embedding, or score-level weighted ensembling) will be explored as an extension of this work while mitigating the increased computational cost.

Finally, the BCI community is increasingly shifting towards open-science and standardisation with a special focus on reproducible pipelines and public data porting beyond the precision of a single number (Ji et al., 2024). In future work, we will align evaluation with these trends by (i) releasing full preprocessing, training scripts and fixed splits, (ii) reporting complementary metrics (Cohen's κ , balanced accuracy F1 under class imbalance, calibration, latency and — in online settings — information transfer rate), and (iii) expanding validation to multi-session and multimodal public benchmarks that stress-test robustness under physiological state shifts. Recent Scientific Data releases of EEG-fNIRS recordings under hypoxia and fatigue-relevant conditions exemplify the field's move towards rigorous, multimodal validation and quality control. Positioning MiniRocket-based feature extraction as a computationally efficient front-end for such benchmarks provides a clear pathway for reproducible clinical-grade assessment.

CRedit authorship contribution statement

Jamal Hwaidi: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mohamed Chahine Ghanem:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Ethics approval and consent to participate

This work reports a secondary analysis of publicly available, de-identified EEG motor-imagery datasets (PhysioNet MI-EEG and BCI-CompIV-2a). The original data collection was conducted in accordance with relevant regulations and institutional guidelines, and participants provided written informed consent, as reported by the dataset curators in Schalk et al. (2004) and Tangermann et al. (2012). The School of Science and Technology, City St George, University of London, Research Ethics Sub-Committee reviewed the present secondary-analysis protocol and determined that it is exempt from ethics approval.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

PhysioNet MI-EEG dataset Schalk et al. (2004) is publicly available at <https://physionet.org/content/eegmmidb/1.0.0/>. BCI-CompIV-2a dataset Tangermann et al. (2012) is publicly available at <https://www.bbc.de/competition/iv/>.

References

- Alomari, M.H., AbuBaker, A., Turani, A., Baniyounes, A.M., Manasreh, A., 2014. EEG mouse: A machine learning-based brain computer interface. *Int. J. Adv. Comput. Sci. Appl.* 5 (4).
- Altaheri, H., Alsulaiman, M., Muhammad, G., 2019. Date fruit classification for robotic harvesting in a natural environment using deep learning. *IEEE Access* 7, 117115–117133.
- Amin, S.U., Alsulaiman, M., Muhammad, G., Mekhtiche, M.A., Hossain, M.S., 2019. Deep learning for EEG motor imagery classification based on multi-layer CNNs feature fusion. *Future Gener. Comput. Syst.* 101, 542–554.
- Ang, K.K., Chin, Z.Y., Wang, C., Guan, C., Zhang, H., 2012. Filter bank common spatial pattern algorithm on BCI competition IV datasets 2a and 2b. *Front. Neurosci.* 6, 39.
- Ang, K.K., Chin, Z.Y., Zhang, H., Guan, C., 2008. Filter bank common spatial pattern (FBCSP) in brain-computer interface. In: 2008 IEEE International Joint Conference on Neural Networks. IEEE World Congress on Computational Intelligence, IEEE, pp. 2390–2397.
- Bengio, Y., Bengio, S., 1999. Modeling high-dimensional discrete data with multi-layer neural networks. *Adv. Neural Inf. Process. Syst.* 12.
- Bengio, Y., Simard, P., Frasconi, P., 1994. Learning long-term dependencies with gradient descent is difficult. *IEEE Trans. Neural Netw.* 5 (2), 157–166.
- Bian, D., Ma, Y., Huang, J., Xu, D., Wang, Z., Cai, S., Wang, J., Hu, N., 2024. Deep-learning-based motor imagery EEG classification by exploiting the functional connectivity of cortical source imaging. *Signal Image Video Process.* 18 (4), 2991–3007.
- Bozhkov, L., Georgieva, P., 2020. Deep learning models for brain machine interfaces. *Ann. Math. Artif. Intell.* 88 (11), 1175–1190.
- Chatterjee, R., Bandyopadhyay, T., 2016. EEG based motor imagery classification using SVM and MLP. In: 2016 2nd International Conference on Computational Intelligence and Networks. CINE, IEEE, pp. 84–89.
- Chen, C.-Y., Wu, C.-W., Lin, C.-T., Chen, S.-A., 2014. A novel classification method for motor imagery based on brain-computer interface. In: 2014 International Joint Conference on Neural Networks. IJCNN, IEEE, pp. 4099–4102.
- de Haan, K., Rivenson, Y., Wu, Y., Ozcan, A., 2019. Deep-learning-based image reconstruction and enhancement in optical microscopy. *Proc. IEEE* 108 (1), 30–50.
- Dempster, A., Petitjean, F., Webb, G.I., 2020. ROCKET: exceptionally fast and accurate time series classification using random convolutional kernels. *Data Min. Knowl. Discov.* 34 (5), 1454–1495.
- Dempster, A., Schmidt, D.F., Webb, G.I., 2021. Minirocket: A very fast (almost) deterministic transform for time series classification. In: Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining. pp. 248–257.
- Derdour, M., Yahiaoui, M.E.B., Kahil, M.S., Gasmi, M., Ghanem, M.C., 2025a. Brain tumour segmentation using choquet integrals and coalition game. *Information* 16 (7), <http://dx.doi.org/10.3390/info16070615>, URL <https://www.mdpi.com/2078-2489/16/7/615>.
- Derdour, M., Yahiaoui, M.E.B., Kahil, M.S., Gasmi, M., Ghanem, M.C., 2025b. Brain tumour segmentation using choquet integrals and coalition game. *Information* 16 (7), 615.
- Devi, S., Singh, K., Singha, N., Singh, M.K., 2024. A comparative analysis for EEG-based motor imagery classification across diverse datasets. In: 2024 First International Conference on Electronics, Communication and Signal Processing. ICECSP, pp. 1–6. <http://dx.doi.org/10.1109/ICECSP61809.2024.10698490>.
- Djamal, E.C., Abdullah, M.Y., Renaldi, F., 2017. Brain computer interface game controlling using fast fourier transform and learning vector quantization. *J. Telecommun. Electron. Comput. Eng. (JTEC)* 9 (2–5), 71–74.
- Dose, H., Möller, J.S., Iversen, H.K., Puthusserypady, S., 2018. An end-to-end deep learning approach to MI-EEG signal classification for BCIs. *Expert Syst. Appl.* 114, 532–542.
- Fenil, E., Manogaran, G., Vivekananda, G., Thanjaivadivel, T., Jeeva, S., Ahilan, A., et al., 2019. Real time violence detection framework for football stadium comprising of big data analysis and deep learning through bidirectional LSTM. *Comput. Netw.* 151, 191–200.
- Fu, R., Tian, Y., Shi, P., Bao, T., 2020. Automatic detection of epileptic seizures in EEG using sparse CSP and fisher linear discrimination analysis algorithm. *J. Med. Syst.* 44 (2), 1–13.
- Ghanem, M., Mouloudi, A., Mourchid, M., 2015. Towards a scientific research based on semantic web. *Procedia Comput. Sci.* 73, 328–335.
- Goldberger, A.L., Amaral, L.A., Glass, L., Hausdorff, J.M., Ivanov, P.C., Mark, R.G., Mietus, J.E., Moody, G.B., Peng, C.-K., Stanley, H.E., 2000. PhysioBank, PhysioToolkit, and PhysioNet: components of a new research resource for complex physiologic signals. *Circulation* 101 (23), e215–e220.
- Graves, A., Schmidhuber, J., 2005. Framewise phoneme classification with bidirectional LSTM and other neural network architectures. *Neural Netw.* 18 (5–6), 602–610.
- Greenfield, L.J., Geyer, J.D., Carney, P.R., 2012. Reading EEGs: A Practical Approach. Lippincott Williams & Wilkins.
- Han, Y., Wang, B., Luo, J., Li, L., Li, X., 2022. A classification method for EEG motor imagery signals based on parallel convolutional neural network. *Biomed. Signal Process. Control.* 71, 103190.
- He, B., Sohrabpour, A., Brown, E., Liu, Z., 2018. Electrophysiological source imaging: a noninvasive window to brain dynamics. *Annu. Rev. Biomed. Eng.* 20, 171.
- Hekmatmanesh, A., Wu, H., Jamaloo, F., Li, M., Handroos, H., 2020. A combination of CSP-based method with soft margin SVM classifier and generalized RBF kernel for imagery-based brain computer interface applications. *Multimedia Tools Appl.* 79 (25), 17521–17549.
- Higashi, H., Tanaka, T., 2012. Simultaneous design of FIR filter banks and spatial patterns for EEG signal classification. *IEEE Trans. Biomed. Eng.* 60 (4), 1100–1110.
- Hou, Y., Jia, S., Lun, X., Hao, Z., Shi, Y., Li, Y., Zeng, R., Lv, J., 2022. GCNs-net: a graph convolutional neural network approach for decoding time-resolved eeg motor imagery signals. *IEEE Trans. Neural Networks Learn. Syst.*
- Hou, Y., Jia, S., Lun, X., Shi, Y., Li, Y., 2020a. Deep feature mining via attention-based bilstm-GCN for human motor imagery recognition. *arXiv preprint arXiv: 2005.00777*.
- Hou, Y., Zhou, L., Jia, S., Lun, X., 2020b. A novel approach of decoding EEG four-class motor imagery tasks via scout ESI and CNN. *J. Neural Eng.* 17 (1), 016048.
- Hu, B., Li, X., Sun, S., Ratcliffe, M., 2016. Attention recognition in EEG-based affective learning research using CFS+ KNN algorithm. *IEEE/ACM Trans. Comput. Biology Bioinform.* 15 (1), 38–45.
- Hyvärinen, A., Ramkumar, P., Parkkonen, L., Hari, R., 2010. Independent component analysis of short-time Fourier transforms for spontaneous EEG/MEG analysis. *NeuroImage* 49 (1), 257–271.
- Jafarifarmand, A., Badamchizadeh, M.A., 2019. EEG artifacts handling in a real practical brain-computer interface controlled vehicle. *IEEE Trans. Neural Syst. Rehabil. Eng.* 27 (6), 1200–1208.
- Ji, X., Bao, B., Li, L.Z., Pu, J., Lin, Y., Zhang, X., Chen, Z., Li, T., 2024. EEG and fNIRS datasets based on Stroop task during two weeks of high-altitude exposure in new immigrants. *Sci. Data* 11 (1), 350.
- Jin, J., Chen, W., Xu, R., Liang, W., Wu, X., He, X., Wang, X., Cichocki, A., 2025. Multiscale spatial-temporal feature fusion neural network for motor imagery brain-computer interfaces. *IEEE J. Biomed. Health Inform.* 29 (1), 198–209. <http://dx.doi.org/10.1109/JBHI.2024.3472097>.
- Jin, J., Fang, H., Daly, I., Xiao, R., Miao, Y., Wang, X., Cichocki, A., 2021. Optimization of model training based on iterative minimum covariance determinant in motor-imagery BCI. *Int. J. Neural Syst.* 31 (07), 2150030.
- Karácsony, T., Hansen, J.P., Iversen, H.K., Puthusserypady, S., 2019. Brain computer interface for neuro-rehabilitation with deep learning classification and virtual reality feedback. In: Proceedings of the 10th Augmented Human International Conference 2019. pp. 1–8.
- Khademi, Z., Ebrahimi, F., Kordy, H.M., 2022. A transfer learning-based CNN and LSTM hybrid deep learning model to classify motor imagery EEG signals. *Comput. Biol. Med.* 143, 105288.

- Lawhern, V.J., Solon, A.J., Waytowich, N.R., Gordon, S.M., Hung, C.P., Lance, B.J., 2018. EEGNet: a compact convolutional neural network for EEG-based brain-computer interfaces. *J. Neural Eng.* 15 (5), 056013.
- LeCun, Y., Bottou, L., Bengio, Y., Haffner, P., 1998. Gradient-based learning applied to document recognition. *Proc. IEEE* 86 (11), 2278–2324.
- Li, H., Ding, M., Zhang, R., Xiu, C., 2022. Motor imagery EEG classification algorithm based on CNN-LSTM feature fusion network. *Biomed. Signal Process. Control.* 72, 103342.
- Li, J., Li, Y., Du, M., 2023. Comparative study of EEG motor imagery classification based on DSCNN and ELM. *Biomed. Signal Process. Control.* 84, 104750.
- Li, J., Li, Y., Yang, H., Du, M., 2022. Research on intention recognition for EEG signals based on DSCNN and GRU. In: *Proceedings of 2021 Chinese Intelligent Systems Conference: Volume I*. Springer, pp. 604–612.
- Li, Y., Yang, H., Li, J., Chen, D., Du, M., 2020. EEG-based intention recognition with deep recurrent-convolution neural network: Performance and channel selection by grad-CAM. *Neurocomputing* 415, 225–233.
- Liao, J.J., Luo, J.J., Yang, T., So, R.Q.Y., Chua, M.C.H., 2020. Effects of local and global spatial patterns in EEG motor-imagery classification using convolutional neural network. *Brain-Computer Interfaces* 7 (3–4), 47–56.
- Liu, Z., Shore, J., Wang, M., Yuan, F., Buss, A., Zhao, X., 2021. A systematic review on hybrid EEG/fNIRS in brain-computer interface. *Biomed. Signal Process. Control.* 68, 102595. <http://dx.doi.org/10.1016/j.bspc.2021.102595>, URL <https://www.sciencedirect.com/science/article/pii/S1746809421001920>.
- Lotte, F., Bougrain, L., Cichocki, A., Clerc, M., Congedo, M., Rakotomamonjy, A., Yger, F., 2018. A review of classification algorithms for EEG-based brain-computer interfaces: a 10 year update. *J. Neural Eng.* 15 (3), 031005.
- Lu, P., Gao, N., Lu, Z., Yang, J., Bai, O., Li, Q., 2019. Combined CNN and LSTM for motor imagery classification. In: *2019 12th International Congress on Image and Signal Processing, BioMedical Engineering and Informatics. CISP-BMEI, IEEE*, pp. 1–6.
- Lun, X., Yu, Z., Chen, T., Wang, F., Hou, Y., 2020. A simplified CNN classification method for MI-EEG via the electrode pairs signals. *Front. Hum. Neurosci.* 14, 338.
- Luo, T.-j., Zhou, C.-l., Chao, F., 2018. Exploring spatial-frequency-sequential relationships for motor imagery classification with recurrent neural network. *BMC Bioinformatics* 19 (1), 1–18.
- Ma, X., Qiu, S., Du, C., Xing, J., He, H., 2018. Improving EEG-based motor imagery classification via spatial and temporal recurrent neural networks. In: *2018 40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society. EMBC, IEEE*, pp. 1903–1906.
- Mahajan, S., Raina, A., Gao, X.-Z., Pandit, A.K., 2022. COVID-19 detection using hybrid deep learning model in chest x-rays images. *Concurr. Comput.: Pr. Exp.* 34 (5), e6747.
- Malan, N., Sharma, S., 2022. Motor imagery EEG spectral-spatial feature optimization using dual-tree complex wavelet and neighbourhood component analysis. *IRBM* 43 (3), 198–209.
- Mathiyazhagan, S., Devasena, M.G., 2025. Motor imagery EEG signal classification using novel deep learning algorithm. *Sci. Rep.* 15 (1), 24539.
- Millán, J.R., Renkens, F., Mourino, J., Gerstner, W., 2004. Noninvasive brain-actuated control of a mobile robot by human EEG. *IEEE Trans. Biomed. Eng.* 51 (6), 1026–1033.
- Moore, M.M., 2003. Real-world applications for brain-computer interface technology. *IEEE Trans. Neural Syst. Rehabil. Eng.* 11 (2), 162–165.
- Muhammad, G., Hossain, M.S., Kumar, N., 2020. EEG-based pathology detection for home health monitoring. *IEEE J. Sel. Areas Commun.* 39 (2), 603–610.
- Muhammad, G., Rahman, S.M.M., Alalaiwi, A., Alamri, A., 2017. Smart health solution integrating IoT and cloud: A case study of voice pathology monitoring. *IEEE Commun. Mag.* 55 (1), 69–73.
- Novi, Q., Guan, C., Dat, T.H., Xue, P., 2007. Sub-band common spatial pattern (SBCSP) for brain-computer interface. In: *2007 3rd International IEEE/EMBS Conference on Neural Engineering. IEEE*, pp. 204–207.
- Olivas-Padilla, B.E., Chacon-Murguia, M.I., 2019. Classification of multiple motor imagery using deep convolutional neural networks and spatial filters. *Appl. Soft Comput.* 75, 461–472.
- Pawar, D., Dhage, S., 2020. Feature extraction methods for electroencephalography based brain-computer interface: A review. *IAENG Int. J. Comput. Sci.* 47 (3).
- Pfurtscheller, G., Da Silva, F.L., 1999. Event-related EEG/MEG synchronization and desynchronization: basic principles. *Clin. Neurophysiol.* 110 (11), 1842–1857.
- Pfurtscheller, G., Neuper, C., 2001. Motor imagery and direct brain-computer communication. *Proc. IEEE* 89 (7), 1123–1134.
- Qamhan, M.A., Altaheri, H., Meftah, A.H., Muhammad, G., Alotaibi, Y.A., 2021. Digital audio forensics: microphone and environment classification using deep learning. *IEEE Access* 9, 62719–62733.
- Ramoser, H., Muller-Gerking, J., Pfurtscheller, G., 2000. Optimal spatial filtering of single trial EEG during imagined hand movement. *IEEE Trans. Rehabil. Eng.* 8 (4), 441–446.
- Reddy, T.K., Arora, V., Kumar, S., Behera, L., Wang, Y.-K., Lin, C.-T., 2019. Electroencephalogram based reaction time prediction with differential phase synchrony representations using co-operative multi-task deep neural networks. *IEEE Trans. Emerg. Top. Comput. Intell.* 3 (5), 369–379.
- Riyad, M., Khalil, M., Adib, A., 2020. Incep-EEGNet: a convnet for motor imagery decoding. In: *International Conference on Image and Signal Processing*. Springer, pp. 103–111.
- Sahu, S.K., Anand, A., 2018. Drug-drug interaction extraction from biomedical texts using long short-term memory network. *J. Biomed. Inform.* 86, 15–24.
- Sakhavi, S., Guan, C., Yan, S., 2018. Learning temporal information for brain-computer interface using convolutional neural networks. *IEEE Trans. Neural Networks Learn. Syst.* 29 (11), 5619–5629.
- Salami, A., Andreu-Perez, J., Gillmeister, H., 2022. EEG-ITNet: An explainable inception temporal convolutional network for motor imagery classification. *IEEE Access* 10, 36672–36685. <http://dx.doi.org/10.1109/ACCESS.2022.3161489>.
- Samek, W., Kawanabe, M., Müller, K.-R., 2013. Divergence-based framework for common spatial patterns algorithms. *IEEE Rev. Biomed. Eng.* 7, 50–72.
- Samek, W., Vidaurre, C., Müller, K.-R., Kawanabe, M., 2012. Stationary common spatial patterns for brain-computer interfacing. *J. Neural Eng.* 9 (2), 026013.
- Schalk, G., McFarland, D.J., Hinterberger, T., Birbaumer, N., Wolpaw, J.R., 2004. BCI2000: a general-purpose brain-computer interface (BCI) system. *IEEE Trans. Biomed. Eng.* 51 (6), 1034–1043.
- Schirrmeyer, R.T., Springenberg, J.T., Fiederer, L.D.J., Glasstetter, M., Eggensperger, K., Tangermann, M., Hutter, F., Burgard, W., Ball, T., 2017. Deep learning with convolutional neural networks for EEG decoding and visualization. *Hum. Brain Mapp.* 38 (11), 5391–5420.
- Shen, Y., Lu, H., Jia, J., 2017. Classification of motor imagery EEG signals with deep learning models. In: *International Conference on Intelligent Science and Big Data Engineering*. Springer, pp. 181–190.
- Sita, J., Nair, G., 2013. Feature extraction and classification of EEG signals for mapping motor area of the brain. In: *2013 International Conference on Control Communication and Computing. ICCCC, IEEE*, pp. 463–468.
- Sun, J., Yan, C., Wen, J., 2017. Intelligent bearing fault diagnosis method combining compressed data acquisition and deep learning. *IEEE Trans. Instrum. Meas.* 67 (1), 185–195.
- Tabar, Y.R., Halici, U., 2016. A novel deep learning approach for classification of EEG motor imagery signals. *J. Neural Eng.* 14 (1), 016003.
- Tangermann, M., Müller, K.-R., Aertsen, A., Birbaumer, N., Braun, C., Brunner, C., Leeb, R., Mehring, C., Müller, K.J., Mueller-Putz, G., et al., 2012. Review of the BCI competition IV. *Front. Neurosci.* 55.
- Tayeb, Z., Fedjaev, J., Ghaboosi, N., Richter, C., Everding, L., Qu, X., Wu, Y., Cheng, G., Conrad, J., 2019. Validating deep neural networks for online decoding of motor imagery movements from EEG signals. *Sensors* 19 (1), 210.
- Thomas, K.P., Guan, C., Lau, C.T., Vinod, A.P., Ang, K.K., 2009. A new discriminative common spatial pattern method for motor imagery brain-computer interfaces. *IEEE Trans. Biomed. Eng.* 56 (11), 2730–2733.
- Tonin, L., Carlson, T., Leeb, R., Millán, J.d.R., 2011. Brain-controlled telepresence robot by motor-disabled people. In: *2011 Annual International Conference of the IEEE Engineering in Medicine and Biology Society. IEEE*, pp. 4227–4230.
- Tortora, S., Ghidoni, S., Chisari, C., Micera, S., Artoni, F., 2020. Deep learning-based BCI for gait decoding from EEG with LSTM recurrent neural network. *J. Neural Eng.* 17 (4), 046011.
- Wang, J., Feng, Z., Lu, N., 2017. Feature extraction by common spatial pattern in frequency domain for motor imagery tasks classification. In: *2017 29th Chinese Control and Decision Conference. CCDC, IEEE*, pp. 5883–5888.
- Wang, Y., Gao, S., Gao, X., 2006. Common spatial pattern method for channel selection in motor imagery based brain-computer interface. In: *2005 IEEE Engineering in Medicine and Biology 27th Annual Conference. IEEE*, pp. 5392–5395.
- Wang, K., Xu, M., Wang, Y., Zhang, S., Chen, L., Ming, D., 2020. Enhance decoding of pre-movement EEG patterns for brain-computer interfaces. *J. Neural Eng.* 17 (1), 016033.
- Wang, K., Zhai, D.-H., Xia, Y., 2019. Motor imagination eeg recognition algorithm based on dwf, CSP and extreme learning machine. In: *2019 Chinese Control Conference. CCC, IEEE*, pp. 4590–4595.
- Wolpaw, J.R., Birbaumer, N., Heetderks, W.J., McFarland, D.J., Peckham, P.H., Schalk, G., Donchin, E., Quatrano, L.A., Robinson, C.J., Vaughan, T.M., et al., 2000. Brain-computer interface technology: a review of the first international meeting. *IEEE Trans. Rehabil. Eng.* 8 (2), 164–173.
- Wu, W., Chen, Z., Gao, X., Li, Y., Brown, E.N., Gao, S., 2014. Probabilistic common spatial patterns for multichannel EEG analysis. *IEEE Trans. Pattern Anal. Mach. Intell.* 37 (3), 639–653.
- Yang, Y., Deng, Q., Shen, F., Zhao, J., Luo, C., 2016. A shapelet learning method for time series classification. In: *2016 IEEE 28th International Conference on Tools with Artificial Intelligence. ICTAI, IEEE*, pp. 423–430.
- Yang, B., Duan, K., Fan, C., Hu, C., Wang, J., 2018. Automatic ocular artifacts removal in EEG using deep learning. *Biomed. Signal Process. Control.* 43, 148–158.
- Yu, Y., Si, X., Hu, C., Zhang, J., 2019. A review of recurrent neural networks: LSTM cells and network architectures. *Neural Comput.* 31 (7), 1235–1270.
- Yuan, H., He, B., 2014. Brain-computer interfaces using sensorimotor rhythms: current state and future perspectives. *IEEE Trans. Biomed. Eng.* 61 (5), 1425–1435.
- Zander, T.O., Kothe, C., 2011. Towards passive brain-computer interfaces: applying brain-computer interface technology to human-machine systems in general. *J. Neural Eng.* 8 (2), 025005.

- Zhang, Y., Nam, C.S., Zhou, G., Jin, J., Wang, X., Cichocki, A., 2018. Temporally constrained sparse group spatial patterns for motor imagery BCI. *IEEE Trans. Cybern.* 49 (9), 3322–3332.
- Zhang, L., Wen, D., Li, C., Zhu, R., 2020. Ensemble classifier based on optimized extreme learning machine for motor imagery classification. *J. Neural Eng.* 17 (2), 026004.
- Zhang, X., Yao, L., Huang, C., Sheng, Q.Z., Wang, X., 2017. Intent recognition in smart living through deep recurrent neural networks. In: *Neural Information Processing: 24th International Conference, ICONIP 2017, Guangzhou, China, November 14-18, 2017, Proceedings, Part II* 24. Springer, pp. 748–758.
- Zhang, R., Zong, Q., Dou, L., Zhao, X., 2019. A novel hybrid deep learning scheme for four-class motor imagery classification. *J. Neural Eng.* 16 (6), 066004.
- Zhang, R., Zong, Q., Dou, L., Zhao, X., Tang, Y., Li, Z., 2021. Hybrid deep neural network using transfer learning for EEG motor imagery decoding. *Biomed. Signal Process. Control.* 63, 102144.
- Zhao, X., Zhang, H., Zhu, G., You, F., Kuang, S., Sun, L., 2019. A multi-branch 3D convolutional neural network for EEG-based motor imagery classification. *IEEE Trans. Neural Syst. Rehabil. Eng.* 27 (10), 2164–2177.
- Zhou, G., Zhao, Q., Zhang, Y., Adali, T., Xie, S., Cichocki, A., 2016. Linked component analysis from matrices to high-order tensors: Applications to biomedical data. *Proc. IEEE* 104 (2), 310–331.